FX Technical Timing

From indicator tests to time-and-currency analysis and economically motivated filters | Final research synthesis | 22 October 2026

A transparent sequence of questions, rules and results

The project began by turning technical indicators into explicit positions. It then asked when those positions work across historical periods and currencies, and finally tested whether economic conditions can reject poor entries or exit deteriorating trades. Every currency-rule account remains an independent time series.

Research layer	What is preserved in this final package
Technical baseline	20 fixed rules; all tested lookback windows; momentum and mean reversion; equal and signal-strength voting; 9 USD currency pairs.
Time and currency	Calendar years and event phases; actual holding paths; daily VIX states; changing relative-rate exposures; commodity and dollar channels.
Economic filters	Earlier technical gates and uniform VIX overrides; two rate-conflict gates; three economic entry hypotheses, their simple OR, and one risk-exit addendum.
Reproducibility	Original workbooks, normalized datasets, frozen results, an executable English notebook, and an offline interactive PnL explorer.

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Unfiltered technical accounts with positive full-history spot returns. This descriptive count motivated deeper diagnosis; it did not select a winning rule for live trading.

The strongest retained historical improvement is CAD Bollinger mean reversion with a commodity/USD entry filter. GBP RSI also improves but is concentrated in a crisis trade. SEK RSI improves mainly through entry filtering; its additional exit contributes only 1.68 percentage points to full-history return.

Evidence status: the main sample, 3 January 2005 to 29 May 2026, was already examined during development. Later filter results are historical research, not a newly untouched OOS. A separate June-October 2026 H.10 replay is retained with its different source and short-sample limitations.

All main comparisons use USD 100,000 initial equity and daily 1x signed spot exposure. Transaction costs, carry, financing and cash interest are excluded as requested. This is a completed research record, with unsuccessful versions retained.

The logical chain

The object of study is a currency pair, a trading rule and a historical state - not a permanent currency style.

Step	Question	Evidence and implication
1. Translate indicators	When exactly are we long, short or flat?	Specify stateful entries, holds and exits; preserve the original indicator windows and voting definitions.
2. Compare fixed rules	Does a rule produce a credible complete account path?	Use a common start, continuous execution, gross daily PnL and reconciled trade ledgers. Keep every rule rather than a rolling winner.
3. Decompose through time	Which years or episodes drive the apparent result?	Inspect inherited positions, long/short contributions, reversals, peak giveback and concentration.
4. Explain currency differences	Why does the same macro period affect different pairs differently?	Combine the USD leg with dated relative rates, commodity income, policy regimes, funding and portfolio channels.
5. Measure trade quality	Is failure about frequency or loss magnitude?	Separate win rate, payoff, average wins/losses, retention, exposure, drawdown and account return.
6. Test economic conditions	Can an observable mechanism improve accepted trades?	Freeze entry gates and a risk-exit experiment; retain all outcomes and compare against the same technical episodes.

What was rejected, and what was carried forward

A rolling-selection workflow was explored and explicitly rejected by the user. It is preserved only as an archive. The final method does not choose a recent best rule, rank currencies, or rotate capital across currencies.

The earliest available uploaded notebook already had delayed execution, compound simple-return accounts and technical exits. Later work did not invent exits or convert a one-day holding strategy into a stateful one. Confirmed refinements include the RSI center-entry guard, a common 2005 start with historical warmup, explicit continuous-account scope, and fuller ledger and data-clock checks.

Original four-mode results and corrected fixed-rule results use different evaluation designs. Their returns are not presented as a like-for-like improvement. All references to an improved filter below compare against its own unchanged technical baseline.

What the literature contributes

Economic and indicator references motivate hypotheses; they do not validate the chosen numerical thresholds.

Source	Relevant insight	Use in this research
Neely, Weller & Ulrich [T1]	Previously studied FX technical rules can lose profitability in later periods.	Motivates fixed-rule temporal validation and an explicit distinction between historical inspection and new data.
John Bollinger [T2]	A band touch alone does not establish a reversal; prices can continue along a band.	Motivates testing mean reversion and breakout as different hypotheses, and examining persistent repricing.
Cboe methodology [T3]	VIX measures option-implied expected US equity volatility over a 30-day horizon.	Uses VIX as a risk-state proxy, not an FX direction signal or a sufficient local-currency risk measure.
Federal Reserve / Curcuru [R1]	Exchange-rate sensitivity to policy expectations varies with currencies, periods and what was already expected.	Motivates relative-rate changes and dollar confirmation, without equating 2Y yield changes to identified policy surprises.
Brunnermeier, Nagel & Pedersen [R2]	Carry exposure can interact with crash risk and position unwinds.	Motivates combining rising risk with the actual higher-yield side of a trade. No claim that VIX reliably forecasts every next FX return.
Central-bank country analysis [R3,R7,R8,C1-C9]	Trade income, monetary policy, funding, portfolio flows and market depth interact.	Provides dated explanations for differences across currencies; avoids permanent momentum/reversion labels.

The fixed cutoffs used here - including VIX levels, rate-spread changes and dollar-index moves - are transparent research conventions. No cited paper establishes those exact thresholds as optimal. Narrative agreement with a historical episode is supporting context, not causal identification.

Sources are linked in the bibliography. Trade profits, counts and curve endpoints are calculated from the supplied/frozen datasets and are separately reproducible in the notebook. The report does not quote long passages from external publications.

Datasets and provenance

Original files, cleaned analysis inputs and frozen outputs are supplied together.

Dataset	Coverage / contents	Role and limitations
FX_Technical_PIT_Dataset(6)(1).xlsx	PIT_Daily_Core: 62,010 rows, 9 currencies x 6,890 dates; 2000-01-03 to 2026-05-29. Also monthly panels, labels, coverage and dictionary.	Canonical FX prices and original context. 2000-2004 initializes indicators; the final continuous accounts begin in 2005. A PIT filename is not proof of complete publication/vintage history.
FX_Features_Book(8).xlsx	11 sheets and 226 profiled fields: spot, rates, growth, inflation, equities, credit, commodities, volatility, positioning, stored technicals and surprises.	Adds economic context, including VIX, 2Y rates, DXY and commodities. It adds no new FX dates; its hash matches the upstream workbook recorded by the earlier PIT file.
Federal Reserve H.10 snapshot	Separate FX source; later-date evaluation 2026-06-01 to 2026-10-16; 96 valid quotes per currency. Historical warmup is separate.	Retained external-source replay of all 20 unchanged rules. H.10 is New York noon, not an identical close or a seamlessly spliced continuation. [D1]
Original notebook and archived designs	Earliest available uploaded notebook; rejected rolling-selection snapshot; fixed/V3-V8 protocols and results.	Preserves provenance and decisions. The original uploaded notebook had no saved outputs; its 792 original-mode account summaries are regenerated in the final notebook.

Data are provided at three levels

data/raw preserves the supplied bytes. **data/clean** provides canonical FX and readable exports of the feature panels. **data/frozen** preserves exact prior research inputs and outputs under their original relative paths. **data/catalog** contains an inventory, field dictionary and SHA -256 manifest.

All nine pairs are studied against USD: EUR, GBP, AUD, NZD, JPY, CHF, CAD, NOK and SEK. FX prices are normalized as USD per unit of foreign currency. This is a set of individual time-series tests; descriptive comparisons across pairs do not create a cross-sectional strategy.

The feature workbook and canonical price panel match at 62,009 of 62,010 aligned price cells. The remaining NOK raw observation on 2024-01-01 is missing in the feature workbook and was already filled in the canonical PIT panel. This discrepancy is retained in the data audit.

Data transformations and information clocks

A plausible economic explanation must be separated from a usable trading input.

Item	Treatment
Price convention	Use P = USD per foreign currency throughout. q=+1 is long foreign / short USD; q=-1 is the reverse. Conventional USD/JPY or USD/CAD quote directions are inverted for this common convention.
Return calculation	Compute simple spot returns directly from canonical prices. Source daily return columns may be log returns; they are not silently substituted into compound simple-return accounts.
Technical inputs	Recompute indicators on canonical prices. Stored workbook technicals can have different native quote directions or RSI smoothing and are not substituted for the fixed strategy definitions.
Economic changes	Compute 5- and 20-observation changes on each native nonmissing series before alignment. They are observation windows, not fixed numbers of calendar days. Long gaps invalidate changes.
E: explanatory state	Latest native observation on or before the return-end date, subject to a 7-calendar-day age limit. Used for contemporaneous attribution only.
K: decision-known convention	Latest native observation strictly before the technical signal date, also no more than 7 calendar days old. Trading uses this convention and a subsequent execution delay.
Missing information	Unknown and stale inputs remain UNKNOWN. A single unknown condition retains the original entry; an OR rule can still reject if another condition is known to be true. Unknown is never classified as low risk.

Two-year yield spreads use local government yields minus US yields, in basis points; German yields proxy the euro area. They are relative-rate-state proxies, not realized carry or directly identified monetary-policy shocks.

The original data do not contain a complete provider publication timestamp, close-timezone and historical-vintage record. Strict prior-date alignment is conservative but does not prove exact point-in-time availability. This remains a real limitation of all economic-filter results.

Feature age caps and gap guards prevent long forward-fills. In particular, sparse NZD early-rate history and NOK rate gaps must remain visible. No later central-bank announcement is backfilled into an earlier trading decision.

Fixed rules: two hypotheses from the same indicator

A window specifies the information horizon; the entry/exit logic specifies the trading hypothesis.

Family / windows	Entry and holding state	Exit
RSI mean reversion 7, 14, 20	Wilder RSI returns from below 30 into [30,50): long foreign. It returns from above 70 into (50,70]: short foreign. Hold the existing state until its exit.	Long exits at RSI >=50; short exits at RSI <=50. The center guard prevents entering after the intended reversion has already completed.
RSI momentum 7, 14, 20	Long when RSI >60; short when RSI <40; flat otherwise. The state is recomputed each signal date.	Return to the neutral band closes the position; crossing to the opposite extreme changes direction with the common lag.
Bollinger mean reversion 10, 20; +/-2 SD	Z=(price-SMA)/sample SD. At Z <=-2, long; at Z >=+2, short. Mean and SD share the chosen window; SD uses ddof=1.	Exit when the position reaches the moving center. A persistent price move can generate a large loss before that happens.
Bollinger breakout 10, 20; +/-2 SD	At Z >=+2, long; at Z <=-2, short. Same price bands, opposite economic hypothesis.	Exit on a return through the moving center.

Mean reversion assumes an observed displacement is likely to reverse sufficiently before losses become large. Momentum assumes directional persistence. Neither hypothesis follows automatically from an indicator reading.

A moving reference can itself shift while the trade is open. For example, a Bollinger trade can exit at its center without earning a positive return: the center may have moved beyond the entry price. The actual trade ledger, rather than an intuitive description of the signal, determines whether it succeeded.

All listed windows remain in the final comparisons. No later result is used to nominate a single permanent RSI or Bollinger window for each currency.

Fixed rules: trend, breakout and combinations

The fixed baseline contains 20 technical rules, two directional references and cash.

Family / windows	Position rule
Moving averages 10/30 and 20/60	Long when the faster SMA exceeds the slower SMA; short when it is below; equality gives a flat target. A change of relationship exits or reverses the position.
Donchian breakout 10 and 20	Use the preceding closes, excluding the current close, for channel extremes. A break above the prior high enters long; a break below the prior low enters short; the channel midpoint governs exit.
MACD 12/26/9	Position is the sign of the difference between the 12/26 EMA spread and its 9-observation signal EMA. Sign changes close or reverse exposure.
Time-series momentum 21 and 63	Position is the sign of the past 21- or 63-observation log price change. Returns are still earned and compounded as simple account returns.
Equal four-signal vote	RSI momentum 14, MA20/60, Donchian 20 and MACD contribute equally. The sign of the sum determines direction; ties are flat.
Strength-weighted vote	The same four signals use fixed deterministic 0-1 price-strength weights with a +/-0.2 threshold. These weights are not estimated win probabilities or recent strategy-performance scores.
Fixed efficiency regime	Prior 63-observation trend efficiency >0.40 selects the MA20/60 signal; <0.20 selects RSI MR14; the middle region is flat. Thresholds are fixed, not fitted.
Benchmarks	Always long foreign, always short foreign, and cash. The short reference is a daily signed exposure, not inverse-quote buy-and-hold.

The four representative rules used for detailed later economic experiments are TSMOM 63, MA20/60, Bollinger MR20 and RSI MR14. They organize two momentum/trend and two mean-reversion hypotheses. The remaining original rules remain available in the baseline notebook and explorer.

Changing the tested window and adding a new economic condition at the same time would confound the explanation. The final filter experiments therefore keep these original technical windows fixed.

Execution, position persistence and account PnL

Signal at close t; execute at the next FX quote close; first new-position return ends at the following quote.

Event	Account interpretation
Technical target	Each indicator determines q in {-1,0,+1}. A signal can maintain an existing position for many days; it is not a mandatory one-day holding.
Execution and held return	executed[t] = target[t-1]; held[t] = executed[t-1]. The account starts in cash at its designated evaluation boundary. Pre-boundary prices initialize signals without creating pre-boundary PnL.
Daily gross return	r_account[t] = held[t] x (P[t]/P[t-1] - 1). Wealth[t] = Wealth[t-1] x (1+r_account[t]). PnL[t] = Wealth[t]-100,000.
Entry gate accepted	Keep the original technical episode and its technical exit. Accepted entries are not recomputed as independent one-day bets.
Entry gate rejected	Stay flat for that entire original episode. Do not re-enter midway when the condition later disappears.
Risk exit	After an accepted entry, the first qualifying risk signal schedules an exit for the next FX close. Retain the return earned up to that close; stay flat for the remainder of the original episode.
Last observation	Open positions are marked to market, not artificially liquidated. A terminal instruction without a later quote cannot be executed early.

Removing a trade changes future account size. Consequently, filtered wealth is rebuilt from its actual retained daily returns; subtracting a rejected trade's old dollar profit from the original final account would be wrong.

All main accounts use daily 1x current-equity spot direction. This is a clearly defined synthetic account convention, not a claim about a specific broker contract, margin arrangement or financing return.

For calendar-period comparisons, the real holding state crosses the boundary and the return segment is rebased to a common capital. A fresh-cash stage experiment is labelled separately. These two conventions answer different questions.

First answer: the full technical-rule map

The entire baseline is retained, including negative results and unselected windows.

All 20 fixed technical rules: full-history return (%)

	EUR	GBP	JPY	CHF	CAD	AUD	NZD	SEK	NOK
RSI_MR_7	+8.6	+16.4	-11.0	+43.1	+40.1	+14.0	+6.3	+8.9	+29.8
RSI_MR_14	-8.7	+15.8	-14.8	+34.2	-7.0	-19.4	+4.4	-2.3	-7.7
RSI_MR_20	+3.5	+17.4	+13.8	+52.9	+2.1	-17.0	-5.7	-24.2	-20.1
RSI_MOM_7	+4.2	-42.5	-24.2	-27.5	-29.9	-0.0	+17.8	-3.4	-43.6
RSI_MOM_14	-13.7	-25.2	-8.9	-32.9	-17.1	-24.4	+1.2	-21.2	-20.9
RSI_MOM_20	-8.9	-24.3	-7.8	-21.5	-5.6	-12.3	-7.1	-27.6	-26.4
MA_10_30	+9.9	-11.9	-1.8	-54.9	-19.7	+0.5	-64.4	-26.1	-24.5
MA_20_60	+10.9	-28.7	+59.2	-47.8	+5.2	-46.4	-36.8	-18.2	-28.7
BB_MR_10	-6.8	+33.4	+10.0	+9.3	+31.4	+135.3	+1.3	+40.0	+39.7
BB_MR_20	+4.3	+37.7	-7.2	+38.5	+5.7	+35.7	-1.0	+32.0	-16.7
BB_BO_10	+0.1	-29.5	-16.6	-19.7	-27.8	-62.3	-12.0	-35.6	-36.9
BB_BO_20	-12.9	-33.6	-4.6	-37.9	-12.7	-38.2	-16.0	-35.1	+0.4
DONCHIAN_10	-6.4	-37.6	-10.4	-25.8	-33.1	-19.3	-2.6	-37.3	-62.1
DONCHIAN_20	+0.1	-42.4	+4.7	-32.8	-29.8	-6.3	-35.0	-20.9	-42.7
MACD_12_26_9	-26.3	+3.3	-6.2	-7.9	-22.1	-51.8	-16.9	+15.0	-41.0
TSMOM_21	+35.9	-25.7	-16.5	-39.3	-24.1	+5.9	-28.7	-11.2	+2.9
TSMOM_63	+15.8	+57.4	+81.8	-55.1	-17.2	-40.0	-33.2	-40.5	-47.4
VOTE_EQUAL_4	-2.1	-43.4	+19.9	-36.9	-25.3	-13.4	-17.3	-12.1	-48.7
VOTE_STRENGTH_4	+3.8	-37.6	-9.4	-48.1	-28.4	-19.0	-30.2	-37.9	-42.8
FIXED_REGIME	+7.8	+2.6	-18.3	+10.4	-5.1	-19.4	+3.4	+32.2	+14.6

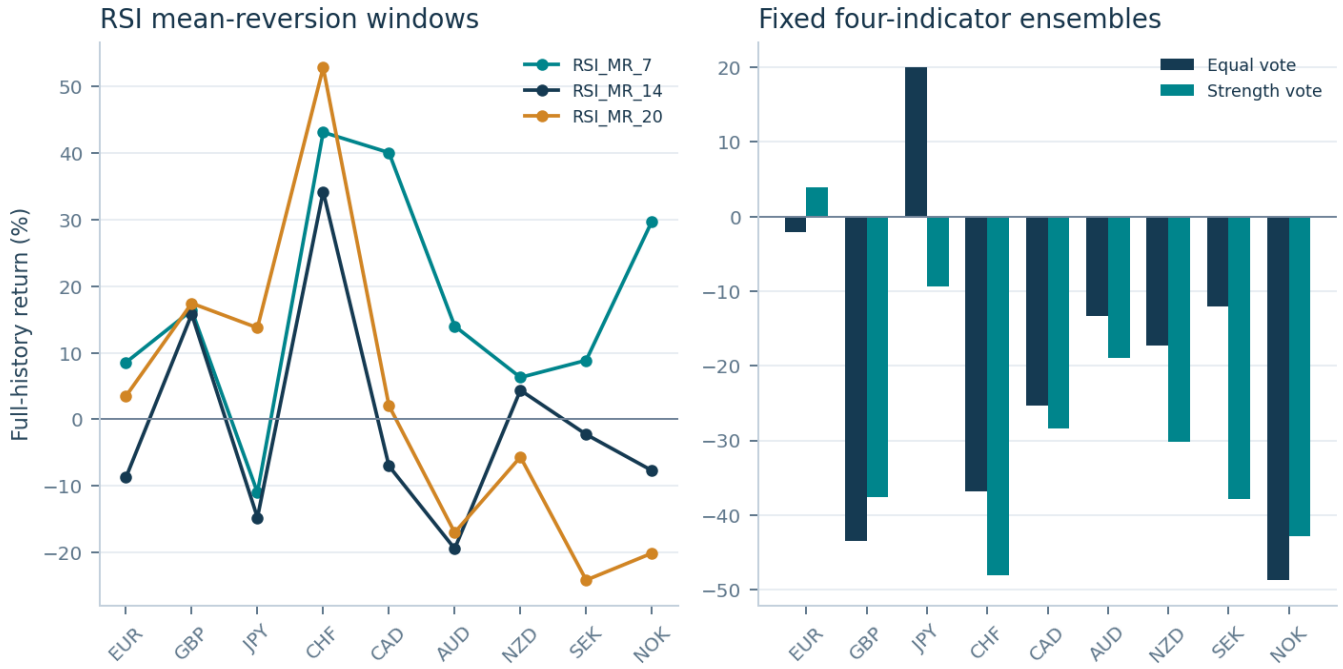
2005-01-03 to 2026-05-29 | Zero costs | Each cell is an independent 1x spot account.
Color saturates at +/-60%; printed values are exact to one decimal.

Only 58 of 180 technical accounts have positive full-history spot returns. Mean-reversion variants are often stronger in this particular sample, but window choice and currency still matter substantially. This map is descriptive historical evidence, not a currency-allocation signal or a list of validated future winners.

The matrix includes 20 rules x 9 currencies. Benchmarks are available separately in the notebook and HTML. Every value represents a continuous independent account from 2005; colored cells are not pooled returns.

Window choice and voting do not solve instability

Keep the different hypotheses visible instead of retrospectively choosing the most attractive curve.



Fixed rule	Positive accounts / 9	Implication
RSI mean reversion 7 / 14 / 20	8 / 3 / 5	Changing the horizon changes signals, trade duration and the full-history result. A good long-period average does not imply uniform performance over time.
Bollinger MR10 / MR20	8 / 6	Both are retained. Their different paths should not be collapsed into a permanent reversion label.
Equal vote / strength vote	1 / 1	Combining correlated price signals does not automatically produce a more robust strategy. Strength weights are not calibrated confidence probabilities.

The window-stability table compares 99 predefined pair/family/window combinations across historical blocks, including position disagreement, daily-return correlation and reversals in the ordering of windows. The notebook preserves every comparison.

The next research question therefore became: what happens to the same unchanged trading rule in different periods and different currencies? This is more informative than treating a full-sample rank as a permanent property.

Intermediate diagnosis: simple technical gates

Earlier controlled experiments remain part of the record; their target-mask logic differs from later episode entry gates.

Earlier hypothesis	Fixed implementation	Interpretation
Bollinger only in a range	BB MR10 or MR20 target allowed when ER63 <0.20; otherwise flat.	Tests the intuition that reversion should work in less directional prices. Re-entry follows the current warmed technical target.
Momentum only in a trend	TSMOM 63 target allowed when ER63 >0.40; otherwise flat.	Tests whether a simple price-efficiency condition improves the timing path.
Momentum and MA confirmation	TSMOM 63 direction is held only when MA20/60 agrees; disagreement gives a flat target.	Checks whether an additional correlated indicator adds useful confirmation.

These V3 controlled variants were exploratory ablations on already examined history. They did not use performance-based rule switching. Their development, validation and later test-labelled segments were cash-reset historical exercises, not newly untouched datasets.

The final notebook retains their specifications and stage results. The explorer exposes the later cash-reset segment as a separate sample so it cannot be accidentally joined to the main continuous account. No new variant or threshold is fitted for this final delivery.

Why proceed to economic conditions?

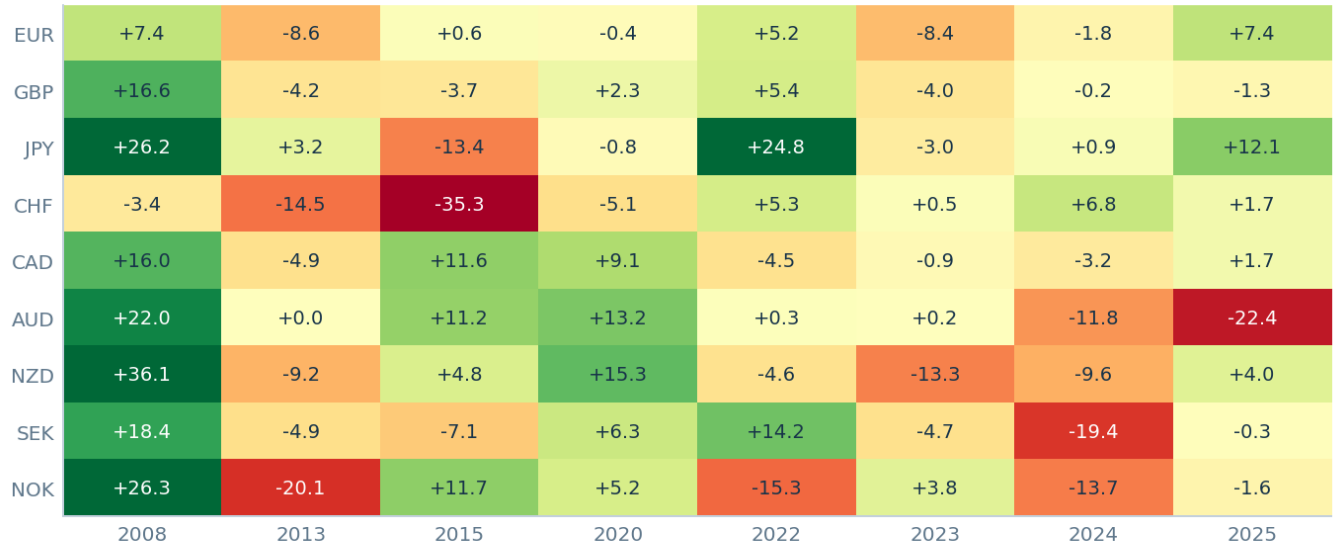
A price-only range label can describe recent path shape without explaining whether the underlying reference value is being repriced. A reversion entry may fail because relative policy, export income, funding needs or portfolio flows continue to push the exchange rate. Those mechanisms motivated the later economic analysis.

That motivation does not imply that an economically named filter must succeed. The same economic condition can also be priced in already, making a rejected entry profitable. Both accepted and rejected trade outcomes must be retained.

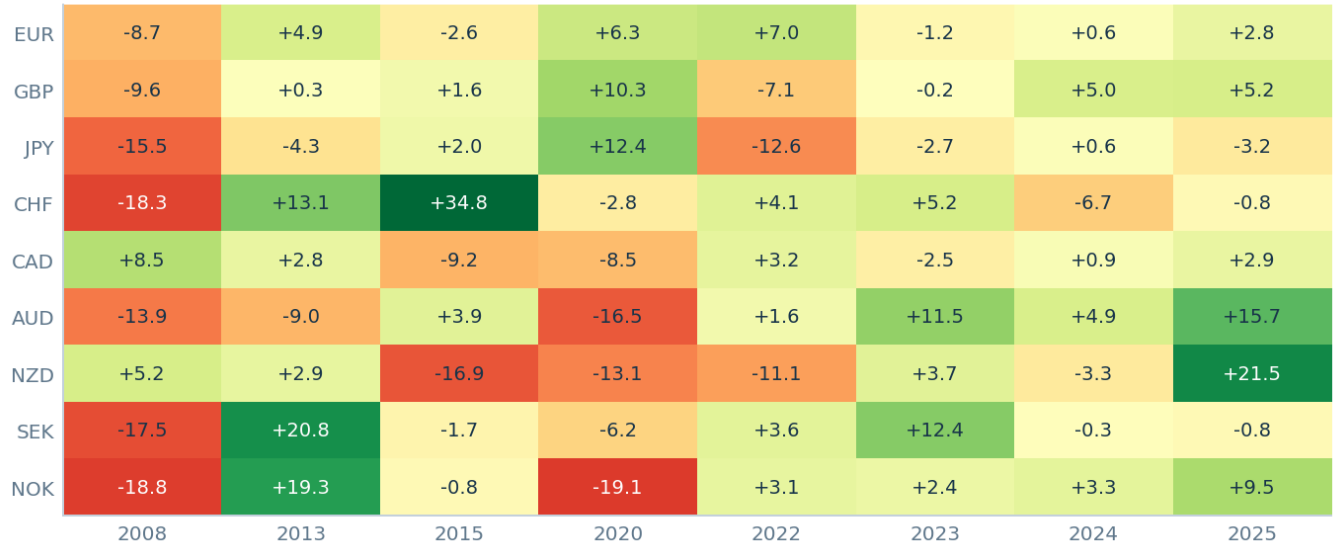
Time x currency: observe the interaction

The same rule changes sign across years; the same year produces different outcomes across currencies.

Moving average 20/60: calendar-year return (%)



Bollinger mean reversion 20: calendar-year return (%)



Example	Calendar-year account return
JPY MA20/60 across years	2015: -13.43%; 2022: +24.82%; 2023: -3.05%
MA20/60 across currencies in 2022	JPY: +24.82%; NOK: -15.25%

The report uses calendar years for comparability, then shorter event phases and actual holdings for explanation. A year is a reporting bucket, not a homogeneous economic regime. All years and event subperiods remain in the tables; the figure shows selected years to make reversals readable.

Currency properties are economic lenses

Structural exposures inform the hypothesis; observable, dated states determine how it is applied.

CCY	Economic lens	Boundary
EUR	Funding and international asset currency; common monetary policy, heterogeneous sovereign risks.	German 2Y yields proxy the euro area; the relative-rate role changes through time. [C1]
GBP	UK-specific policy and institutional uncertainty interact with global portfolio positions.	A valuation change in external assets is not automatically a capital flow or a reversal signal. [C2]
AUD	Commodity income and terms of trade interact with relative rates and global risk.	Copper is only a partial proxy; demand and supply shocks can differ. [C3]
NZD	Dairy and other soft-commodity exports, domestic rates and global risk.	No dairy series is supplied; do not substitute copper or oil mechanically. [C4]
JPY	Funding exposure, policy regimes and international position adjustments.	Risk-off appreciation is not proof of repatriation; low rates are not a permanent momentum instruction. [C5]
CHF	Safe-haven exposure conditioned by intervention and exchange-rate arrangements.	The 2015 floor removal concerned EUR/CHF, not a USD/CHF peg. [C6]
CAD	Resource income and the common US-dollar factor jointly matter.	CAD is not an oil contract; Brent and DXY do not identify a unique structural shock. [C7]
NOK	Energy income, overseas assets, hedging, fiscal conversion and market depth.	Oil receipts need not become immediate NOK purchases; rate data have long gaps. [C8]
SEK	Portfolio flows, relative policy expectations and changing risk premia.	The higher-yield side can switch between SEK and USD; use the actual trade direction. [C9]

Every exchange rate here has two legs. A common dollar funding shock can dominate a foreign currency's usual commodity or safe-haven behavior. Conversely, a US-specific asset revaluation can weaken the dollar during rising risk. The sign must be read in the specific episode, not imposed by a currency label.

The accompanying currency_economic_lenses table contains the dated cases, numerical rate examples and primary links behind this summary. These observations motivate state variables; they do not assign a permanent strategy to any currency.

The same currency can change its role

Current relative rates and the trade direction matter more than a timeless high-yield label.

Currency roles change: median local minus US 2Y yield (bp)

	2005	2008	2012	2015	2020	2022	2025
EUR	-156.1	+157.1	-21.0	-88.0	-89.5	-231.8	-189.6
GBP	+40.6	+218.2	+1.0	-10.9	-21.5	-94.8	+16.5
JPY	-367.9	-146.1	-16.8	-64.7	-32.0	-313.1	-308.2
CHF	NA	+1.0	-47.7	-156.1	-97.1	-282.8	-385.6
CAD	-66.0	+64.8	+83.1	-12.7	+11.6	-1.6	-118.2
AUD	+146.6	+399.5	+247.2	+123.1	+1.0	-43.0	-35.1
NZD	+248.5	+453.5*	NA	+214.1	-1.9	+51.6	-62.0
SEK	NA	+182.8	+59.3	-99.8	-52.6	-154.2	-192.1
NOK	NA	NA	+105.2*	-0.7	+4.3*	-30.4	-14.5

* Less than 50% rate coverage: NZD 2008 11/262; NOK 2012 25/261; NOK 2020 112/262.
NA = unavailable. Sparse annual medians should not be read as representative full-year exposures.

AUD's median known 2Y spread versus the US changes from about +399.5bp in 2008 to -35.1bp in 2025. SEK changes from +182.8bp to -192.1bp. A long foreign position and a short foreign position therefore carry different relative-rate exposures in different periods.

For q=+1 (long foreign), q x spread is positive when the foreign currency is the higher-yield side. For q=-1 (short foreign / long USD), the same signed calculation is positive when USD is the higher-yield side. This permits currency roles to change without rewriting the rule by currency or year.

Coverage changes what can be concluded

Some annual medians are based on very few observations. NZD 2008 has only 11 known-rate days out of 262; NOK has no 2008 rate coverage and a long 2020 gap. These values are flagged, not used to manufacture a complete historical rate narrative.

Market 2Y yields are a practical relative-policy and risk-state proxy. They are not overnight interest earned on the FX position, and the account does not include carry. [R1,R2]

Risk states: VIX level, direction and timing

A high VIX reading does not identify the direction of the next currency move.

Component	Fixed definition	Use
Level	Low: VIX <20; middle: 20 to below 30; high: >=30.	A transparent state partition, not an official Cboe crisis classification.
Change	Up: 5 native observations rise >=3 points; down: fall >=3; otherwise stable.	Separates rising pressure from easing pressure at a similar level.
Nine states	Cross three levels with three directions; preserve UNKNOWN.	Report daily mean returns, active days, exposure side, episodes and year coverage.
Panic episode	Starts at VIX>=30; ends after 10 consecutive native observations below 25; following 20 observations describe recovery.	A completed episode and its peak are retrospective labels, never available early for trading.
Local FX risk	Country FX implied volatility, where available, plus local policy events.	Global equity risk does not capture every FX-specific discontinuity.

Contemporaneous state attribution sums original daily dollar PnL inside each state and reconciles to the original account. It explains where profits happened. It is not a trading strategy obtained by compounding only favorable-state days.

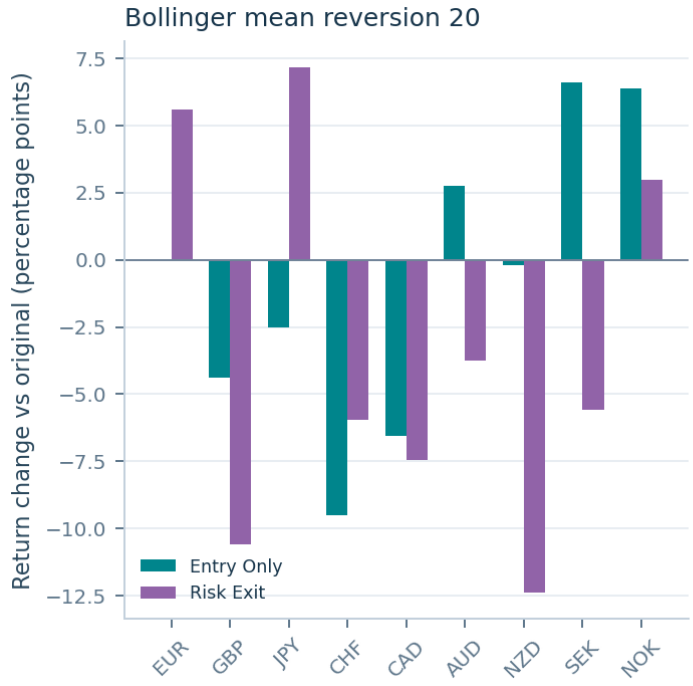
Entry decisions instead use the prior-date K state and the common execution delay. The original technical episode is accepted or rejected as a whole. The distinction between E attribution and K decisions is maintained in the notebook.

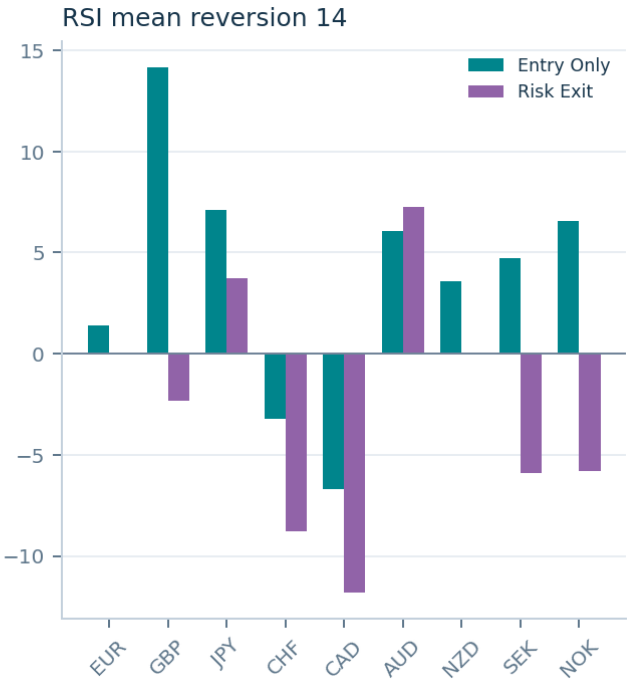
Cboe defines VIX from S&P 500 option prices as expected 30-day US equity volatility. This supports a broad risk lens, not a claim that VIX is an FX forecasting model. [T3]

The daily feature panel also includes relative rates, DXY, commodity context and local FX volatility. Some variables are descriptive only; the final economic rules use a deliberately small subset.

A uniform panic switch was not sufficient

The first VIX intervention used the same stress rule across all currencies and both core mean-reversion rules.





V5 intervention	Bollinger MR20: return improved	RSI MR14: return improved
Reject new entries when VIX>=30 and 5-observation rise>=3	3 / 9	7 / 9
Same entry gate plus an early risk exit	3 / 9	4 / 9

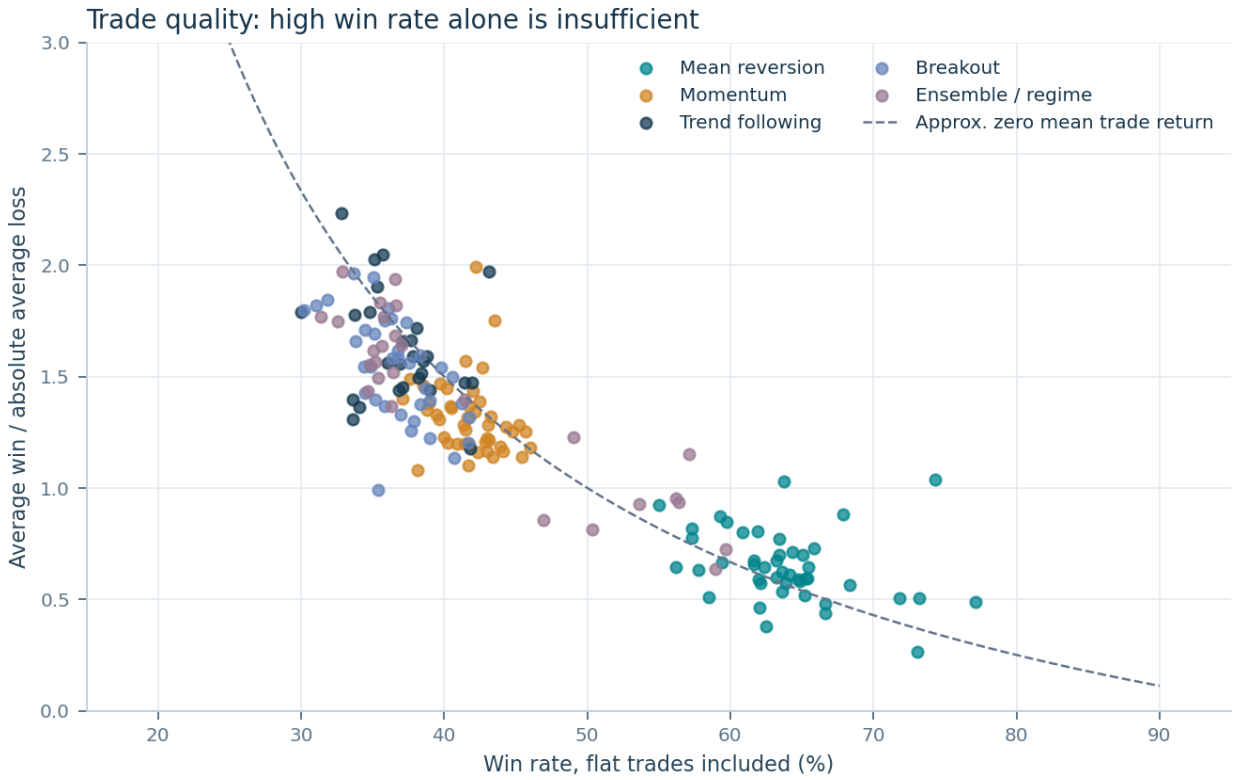
Drawdowns often improve, but improved capital protection is not identical to improved final return. A rule can exit after a sharp adverse move and miss the subsequent recovery. The effects also differ between a Bollinger entry at an extreme and an RSI entry after returning from an extreme.

This result provided the bridge to the later filters: add the side of the trade and a measurable currency exposure, rather than using a common VIX level as a universal on/off instruction.

The V5 threshold of VIX>=30 is distinct from the later V8 risk/yield condition of VIX>=20 plus rising pressure and signed yield exposure. Both versions remain available; their different definitions are not silently merged.

Trade quality: frequency and magnitude

The user's next objective was to improve win rate and payoff, while preserving a credible account PnL.



Measure	Definition / interpretation
Win rate	Winning closed trades / all eligible closed trades, including flats. Open trades remain separate.
Payoff ratio	Mean winning whole-trade return / absolute mean losing whole-trade return. Undefined if the required cohort is missing.
Expected trade return	Win probability x mean win minus loss probability x mean absolute loss; flat trades have zero contribution.
Other diagnostics	Average holding time, trade retention, exposure, return profit factor, dollar profit factor, account return, drawdown and concentration.

The baseline ledger contains 54,262 technical entries: 54,187 eligible closed trades and 75 open trades. These are overlapping rule accounts, not 54,262 independent statistical observations. A daily hit rate is not the same as the win rate of the full technical episode.

An entry-year cohort can contain trades that finish later. Calendar-year PnL instead records returns when they occur, including positions inherited from the previous year. Both views are supplied and clearly distinguished.

Economic filters: exact frozen conditions

Let q=+1 mean long foreign, q=-1 long USD; d is local 2Y minus US2Y yield in basis points.

Version	Reject a new technical entry when...	Mechanism
V7 rate change	q x change 20(d) <= -10bp.	Recent relative-rate repricing opposes the intended position.
V7 rate level + change	The preceding condition AND q x d <= -50bp.	Both current relative-rate level and its change conflict with the position.
V8 policy + USD	q x change 20(d) <= -10bp AND q x DXY20 return >=1%.	Relative-rate repricing and broad-dollar direction jointly oppose the trade.
V8 risk + yield	VIX>=20 AND change 5(VIX)>=3 points AND q x d >=50bp.	Rising risk pressure while buying the relatively higher-yield currency.
V8 commodity + USD	q x commodity 20 return <0 AND q x DXY20 return >=1%. Only AUD/CAD/NOK.	Commodity income and dollar direction jointly oppose the position. AUD uses copper; CAD/NOK use Brent.
V8 any economic gate	Any known policy/USD, risk/yield or applicable commodity/USD condition is true.	A simple OR frozen with the individual hypotheses before their new results were viewed.
V8 risk gate + exit	Entry rule equals risk/yield. After entry, first occurrence of the same condition schedules next-close exit.	Tests pressure emerging during the holding period; no mid-episode re-entry.

DXY mechanically overlaps some studied currencies, especially EUR. It is a correlated price confirmation, not an independent exogenous macro factor. Copper and Brent are limited proxies, and the absence of a dairy series prevents a comparable NZD commodity gate. [R3-R5,R7,R8]

All applicable conditions use the K information convention. Missing inputs do not automatically reject a trade; a known rejection by another OR component can still do so. The original technical entry/exit logic remains fixed.

Hypothesis, validation and retained results

Three research rounds were recorded; no thresholds were adjusted after the later V8 review.

Round	What changed	What the result led us to inspect
1. Separate mechanisms	Policy/USD, risk/yield and commodity/USD entry gates, each fixed before its new outcomes.	Whether gains come from fewer losing entries, reduced loss magnitude, lower exposure or missed winners.
2. Simple combination	A pre-specified OR of the same three conditions, not a combination assembled after seeing their profits.	Broader PnL and drawdown effects, but insufficient joint win/payoff improvement and sometimes low trade retention.
3. Holding-period pressure	After development-period diagnosis, freeze one risk-exit addendum before reviewing its later-period candidate outcomes.	Distinguish entry improvements from the incremental effect of exits; retain successful and harmful early exits.

Meaningful historical improvement screen

Full history: win rate improves by at least 2 percentage points; payoff improves relatively by at least 10%; total return improves by at least 5 percentage points; drawdown worsens by no more than 2 points; at least 50% of closed trades are retained; at least 30 trades close.

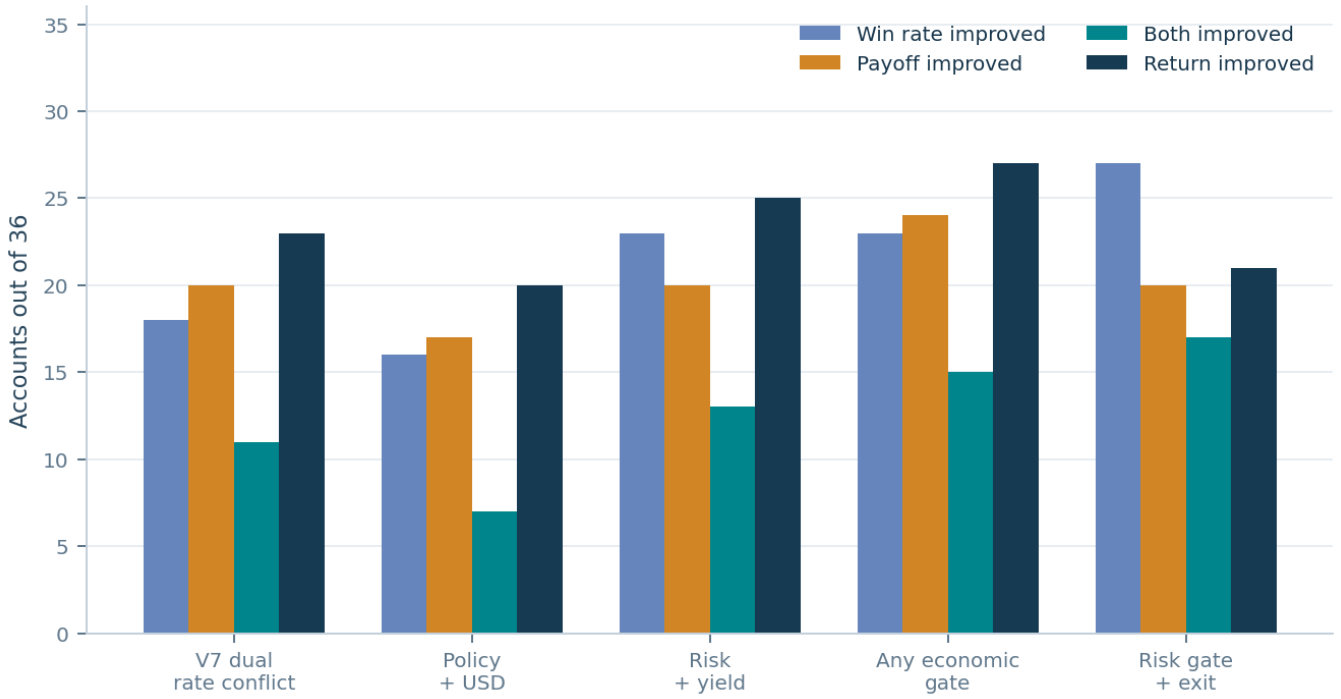
Later historical review (2015 onward): win rate, payoff and return all improve; drawdown worsens by no more than 2 points; at least 20 trades close. At least 3 of 5 fixed calendar blocks must show a return improvement.

Five variant rows pass, representing only three original accounts. GBP policy/USD and OR have identical paths for that RSI account; CAD has two related passing filter versions. These are not five independent discoveries.

Development/review ordering constrains the current iteration, but earlier analysis had already inspected the underlying history. Passing this screen is a historical candidate result, not statistical significance, a causal result or newly pristine OOS.

Breadth: improvements are not universal

The denominator is 36 core accounts, except the commodity gate, which has 12 applicable accounts.



Filter	Win rate up	Payoff up	Both up	Return up	DD no worse
Rate level + change	18/36	20/36	11/36	23/36	22/36
Policy + USD	16/36	17/36	7/36	20/36	28/36
Risk + yield	23/36	20/36	13/36	25/36	26/36
Commodity + USD	9/12	11/12	8/12	11/12	10/12
Any economic gate	23/36	24/36	15/36	27/36	32/36
Risk gate + exit	27/36	20/36	17/36	21/36	28/36

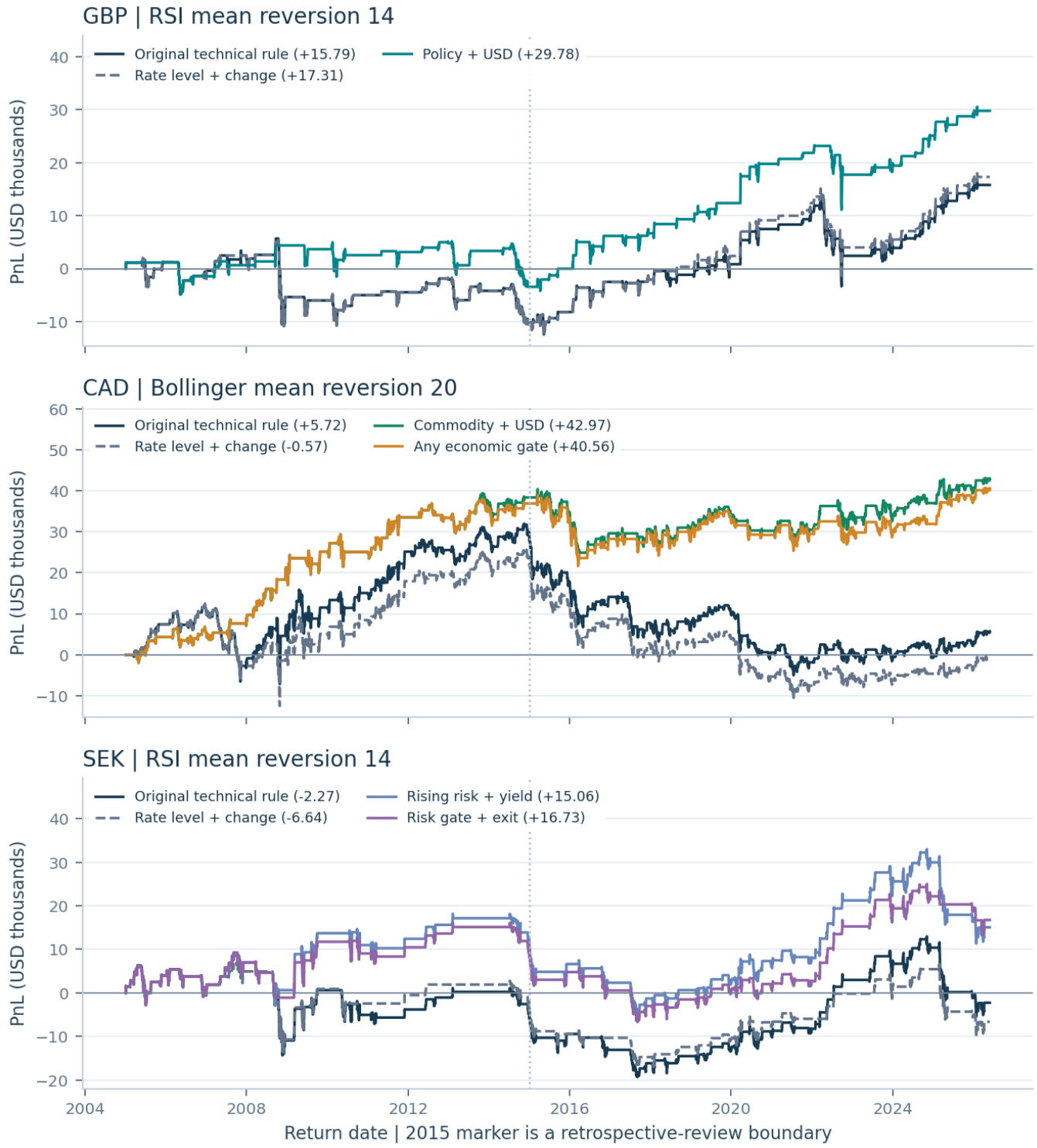
No general filter satisfies all pre-specified broad-coverage targets together: at least 18/36 accounts with both quality measures higher, at least 24/36 with higher return, and at least 24/36 with no worse drawdown.

Risk exits increase the win rate of 27/36 accounts relative to the original strategy, but improve return in only 21/36. Relative to risk entry filtering alone, adding the exit improves return in only 16/36. A higher win rate does not establish that an exit is broadly beneficial.

The commodity gate is not applicable to the other 24 accounts. Identity copies outside AUD/CAD/NOK are not counted as successful filters. All full, development and later-review counts are supplied.

Three historical improvements: complete PnL

Full continuous accounts; the 2015 marker identifies the retrospective-review boundary, not a new cash restart.



Initial capital USD 100,000 per independent account. Legend endpoints are PnL in USD thousands. All fees, carry and financing are excluded. Gains can persist alongside long flat periods and substantial drawdowns.

Quality improved, with different mechanisms

The same results are shown relative to both the original strategy and the earlier V7 reference.

Account	Version	Closed	Win rate	Payoff	Return	Max DD
CAD Bollinger MR20	Original	185	64.86%	0.583	+5.72%	28.01%
CAD Bollinger MR20	Rate level + change	173	64.74%	0.557	-0.57%	28.85%
CAD Bollinger MR20	Commodity + USD	147	67.35%	0.786	+42.97%	12.44%
GBP RSI MR14	Original	78	71.79%	0.506	+15.79%	17.11%
GBP RSI MR14	Rate level + change	74	72.97%	0.491	+17.31%	17.11%
GBP RSI MR14	Policy + USD	57	77.19%	0.619	+29.78%	9.87%
SEK RSI MR14	Original	72	63.89%	0.572	-2.27%	26.17%
SEK RSI MR14	Rate level + change	53	64.15%	0.526	-6.64%	22.53%
SEK RSI MR14	Risk gate + exit	69	66.67%	0.656	+16.73%	19.62%

CAD: mean winning trade return decreases slightly from 1.023% to 0.977%, but mean loss magnitude falls from 1.756% to 1.244%. The improvement primarily comes from the loss side. Of 38 rejected trades, 21 would have won and 17 would have lost.

GBP:21 entries are rejected: 12 original winners,8 losers and 1 flat. Better outcomes do not require most rejected trades to have been losers; their magnitudes and timing matter.

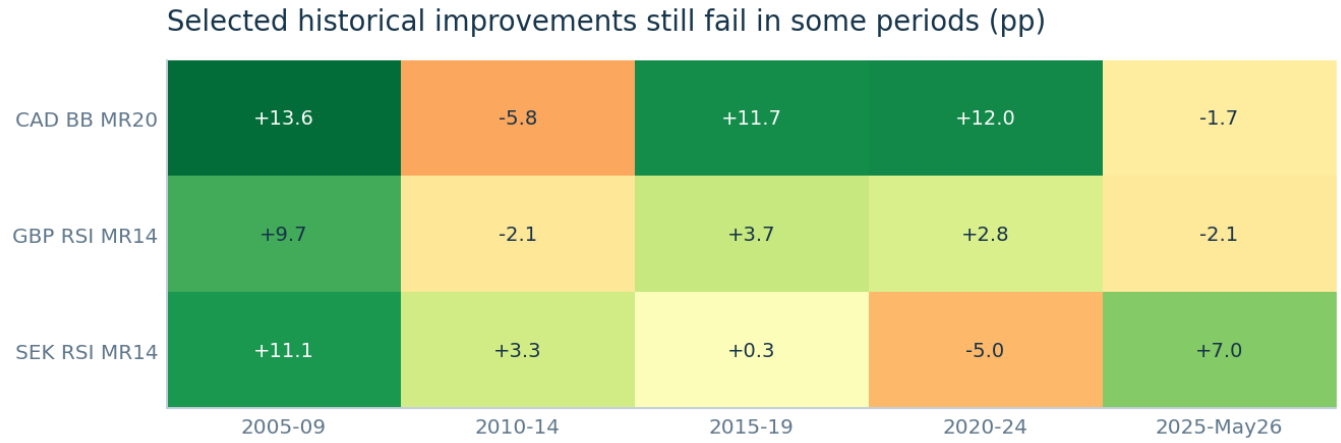
SEK: entry filtering alone produces 15.06% full-history return; adding exits produces 16.73%. The extra exit contributes 1.68 percentage points, with the same 66.67% win rate. Most of the full improvement must not be attributed to the exit.

Absolute returns remain modest. The retained CAD, GBP and SEK variants have calendar CAGRs of about 1.68%,1.23% and 0.73%, respectively, before costs. Improved historical quality is not equivalent to a high-return live trading system.

The improvement still depends on the period

Calendar blocks preserve the continuous holding state. Later-review quality uses entries within that review and closed by its end.

2015-2026 review	Win rate	Payoff	Period return
CAD Bollinger MR20	60.00% to 63.95%	0.531 to 0.619	-18.52% to +3.34%
GBP RSI MR14	78.72% to 86.11%	0.704 to 0.978	+29.34% to +34.39%
SEK RSI MR14	66.67% to 68.42%	0.661 to 0.724	+2.14% to +6.23%



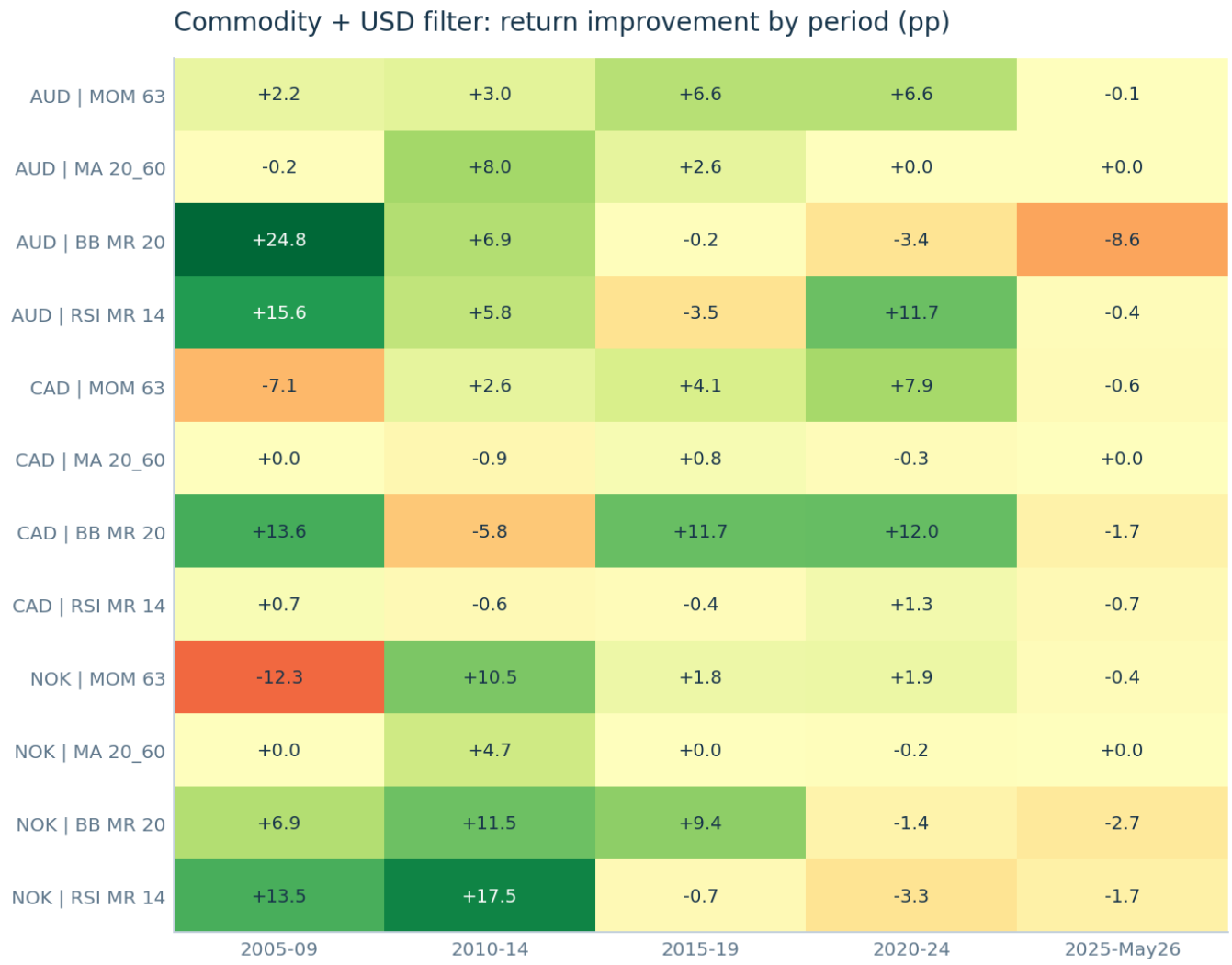
CAD and GBP are both worse in 2010-2014 and 2025-May 2026 despite higher full-history returns. SEK is worse in 2020-2024 and better in the latest block. The result belongs to a rule, a currency and a state-dependent path; it is not a permanent currency property.

The review-period return includes returns from inherited positions. Its win rate and payoff use the new-entry closed cohort. This is why a period PnL table and an entry-cohort table should not be treated as interchangeable.

The late data were already seen during the research sequence. This page checks historical consistency; it does not supply a fresh OOS performance claim.

Commodity currencies do not move as one group

All 12 applicable currency-rule accounts and all 5 calendar blocks are retained.



AUD Bollinger MR20 and NOK RSI MR14 are hurt by the commodity/USD gate in each of the last three blocks. Their strong earlier contribution can conceal this deterioration in a full-history total. CAD Bollinger has a different pattern, with improvements in 2015-2019 and 2020-2024.

A NOK MA20/60 long entered on 30 March 2022 when Brent's 20-observation change was +11.38%, yet the original trade lost 9.03%. The commodity gate accepted it; the policy/USD gate rejected it because the relative-rate change was -28.7bp and DXY was +2.47%.

Energy income, relative rates, dollar demand and portfolio/hedging channels can offset one another. Norges Bank's 2022 discussion supports a multi-channel explanation rather than an automatic oil-to-krone link. [R8]

CAD: a low-VIX reversion failure

The strongest retained historical candidate avoids some sustained repricing episodes, while still rejecting profitable reversals.

Original entry	Trade direction	Known Brent 20	Known DXY 20	Known VIX	Original trade
2007-03-22	Short CAD	+3.83%	-1.32%	13.27	-8.31%
2014-12-09	Long CAD	-17.17%	+1.93%	11.82	-8.18%
2011-11-22	Long CAD	-1.83%	+2.18%	32.00	+2.41%

The December 2014 Bollinger long remained open until 16 February 2015 and lost 8.18%. At the original entry, falling oil and a stronger dollar opposed the direction even though VIX was only 11.82. A panic-only gate would not capture this mechanism.

A plausible interpretation is sustained adjustment in Canadian income, terms of trade and policy expectations, rather than a short-lived displacement around an unchanged mean. The Bank of Canada's 21 January 2015 rate cut explicitly addressed the oil-price shock. It occurred after entry and is used only as holding-period context, not a forecast input. [R6]

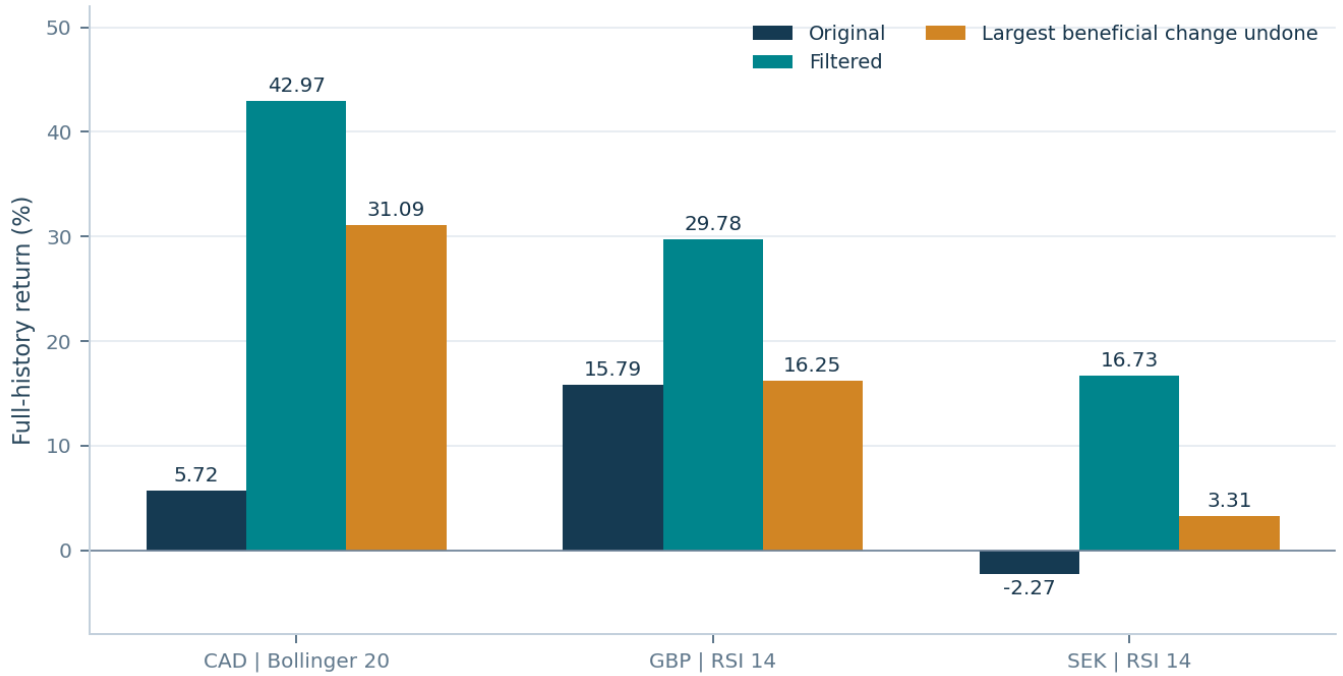
The 2007 short-CAD example shows that the rule is direction-aware: commodity strength and dollar weakness opposed selling CAD, and the gate avoided another substantial loss. This is not merely a restriction on long CAD entries.

The 2011 long would have earned 2.41%, and another rejected August 2010 long would have earned 2.34%. Recent commodity and dollar moves can already embody the shock and precede recovery. The 20-observation gate cannot identify that distinction reliably.

Bank of Canada work also emphasizes common dollar factors. The joint condition has a stronger economic rationale than an oil-only narrative, but related price factors do not establish causality. [R3]

How much depends on one favorable change?

Undo the largest beneficial episode modification, then recompute compound wealth.



CAD: the filtered return falls from 42.97% to 31.09% when its largest beneficial change is undone. It remains well above the original 5.72%. This supports a less concentrated return improvement, not a guarantee about other metrics or future performance.

GBP: undoing the largest favorable change reduces 29.78% to 16.25%, close to the original 15.79%. A long entered 14 October 2008 and closed 16 December would have lost 10.43%; rejecting it is central to the apparent benefit.

The GBP entry had a known 20-observation spread change of-44.13bp, DXY +5.10%, and VIX 69.95. The policy and risk conditions both opposed the trade. The mechanism is plausible, but the result is not as broadly supported as its headline return suggests.

A2022 GBP example is also consistent: the 10 March long lost 4.23% by its May exit, while relative-rate change and dollar direction opposed entry. The September mini-budget cannot explain a trade that ended in May. Rejected profitable shorts in 2010 and 2025 remain in the opportunity-cost table.

Largest-episode sensitivity is an ex post concentration diagnostic. It does not create a strategy that knows which single trade to remove.

SEK: the higher-yield side can be USD

A direction-aware risk condition permits the same currency to play different roles in different periods.

Date / decision	Available state	Result
24 February 2025 entry	Short SEK / long USD; known spread about -226bp; VIX 15.66.	Risk entry condition not triggered; original RSI trade accepted.
28 February risk signal	Uses 27 February data: VIX 21.13,5-observation rise 5.47 points, spread -211.1bp. q=-1, so signed spread +211.1bp.	The risk/yield exit condition becomes true.
3 March execution	Exit at the next FX close, including the return earned through that close.	Trade return -1.52%, versus -9.25% under the original technical exit on 13 May.

The risk condition concerns buying the relatively higher-yield currency; in this 2025 trade that currency was USD. In 2008 the gate instead rejected a long-SEK trade and avoided an11.50% loss. No permanent high-yield or safe-haven label is required.

Riksbank material published in March 2025 discusses krona strength alongside US asset outflows, European asset flows and changing relative expectations. It helps explain the observed path but was published after the exit; it was not available for the February risk signal. [R9,R10]

The opposite example occurred in March 2020: a short-SEK/long-USD trade would have earned 1.93%, but risk exit left only about 0.014%. Dollar funding demand during that crisis provided a different environment. BIS discusses this dollar appreciation and financing pressure. [R12]

The extra SEK exit improves full-history return by only 1.68 percentage points beyond entry filtering. A compelling individual 2025 example should not be mistaken for proof that exits account for the whole improvement.

Risk exits: tail protection versus lost recovery

The information clock and the post-exit price path determine whether the same rule helps.

Original JPY short	Original technical exit	Risk exit	Risk signal / execution
Bollinger, entered 2 September 2008	-10.858%	-0.324%	5 September /8 September
Bollinger, entered 18 July 2024	-6.865%	-8.398%	5 August /6 August
RSI, entered 30 July 2024	-5.487%	-5.613%	5 August /6 August

The 2008 exit avoided substantial subsequent losses. In 2024, the same risk rule exited after the sharp move and missed part of the later recovery. On 5 August it used the 2 August VIX observation of 23.39, not the higher 5 August close, which was not yet in its known input set.

BIS describes a macro-news trigger amplified by leveraged unwinds and margin dynamics, followed by relatively rapid stabilization in August 2024. That account is consistent with the risk of exiting late in a reversal, but does not establish a predictable turning point. [R11]

Among 554 comparable original closed trades	Count
Original loss becomes profit	100
Original profit becomes loss	62
Still a loss, but smaller	108
Still profitable, but less profit	121
Higher / lower trade return overall, including other cases	302 /252

There are 556 early exits in total; 2 original episodes remain open at cutoff and are compared by marked-to-market value separately. Momentum/trend accounts have 86 loss-to-profit versus 29 profit-to-loss conversions; mean reversion has 14 versus 33. This illustrates why a common protective exit can remove recoveries central to reversion trades.

External dates and source sensitivity

The project contains a later-date technical replay, but no new untouched economic-filter OOS record.

Evidence set	What was done	What it establishes
Main workbook history	All 20 fixed rules,2005-01-03 to 2026-05-29; prior prices for warmup.	A reproducible historical comparison. Repeated inspection prevents later variants from claiming a pristine holdout.
Separate H.10 extension	2026-06-01 to 2026-10-16;96 valid quotes per pair; all 20 rules unchanged; no source splice.	114/180 technical accounts positive in a short retrospectively downloaded extension. Different quote time and calendar prevent a source-identical continuation claim.
Source/calendar overlap check	2024-01-02 to 2026-05-29: A=original quote/calendar; B=original quotes on H.10 valid dates; C=H.10 quotes/calendar.	A/B isolates calendar changes; B/C isolates quote changes on a common calendar. Not new OOS.
V7/V8 economic variants	The same already-seen main history plus an internal development/review split.	Historical candidate evaluation. The matching new macro panel stops in May 2026, so the later H.10 prices alone cannot validate these filters.

Calendar changes alone flip the full-period PnL sign in 20/180 overlap accounts; changing quote source on the same valid dates flips 30/180. Changing both flips 34/180. Observation-count windows and execution opportunities are sensitive to the quote calendar, even without changing rule parameters.

H.10 contains New York noon buying rates and removes non-quote ND dates in the retained replay. Historical releases and current downloadable histories can differ after revisions. This is why the exact snapshot and source-sensitive comparisons are supplied. [D1]

The word OOS is reserved here for a test with a defensible information and design boundary. Renaming already examined dates or reusing the observed H.10 outcomes for new filter design cannot regenerate an untouched validation sample.

What the completed research supports

The final state is a transparent research package, with strengths and failures visible.

Conclusion	Evidence
Keep the baseline broad	The 20 original rules and windows reveal substantial differences; only 58/180 original technical accounts are positive. A single full-history winner is an inadequate summary.
Use time and currency together	The same rule changes sign across periods, and currencies react differently within the same period. Currency traits are economic context, not permanent style assignments.
Economic entry filters can help	CAD Bollinger plus commodity/USD filtering improves win rate, payoff, return and drawdown, with a less concentrated gain than GBP.
Distinguish entry from exit	SEK gains mainly come from rejecting entries; its added exit is a smaller incremental improvement. JPY and 2020 SEK provide concrete harmful-exit counterexamples.
Retain evidence boundaries	No common filter meets all broad targets. The latest filters are not fresh OOS; exact publication-time data and a genuinely new test record remain absent.

How to use the delivered files

Notebook: read the research sequence and run the executable reconstruction. It rebuilds the original-mode summaries, all fixed-rule baseline accounts, the macro features and the frozen economic engine, then reconciles outputs to the archived results. Historical intermediate stages are retained as labelled evidence.

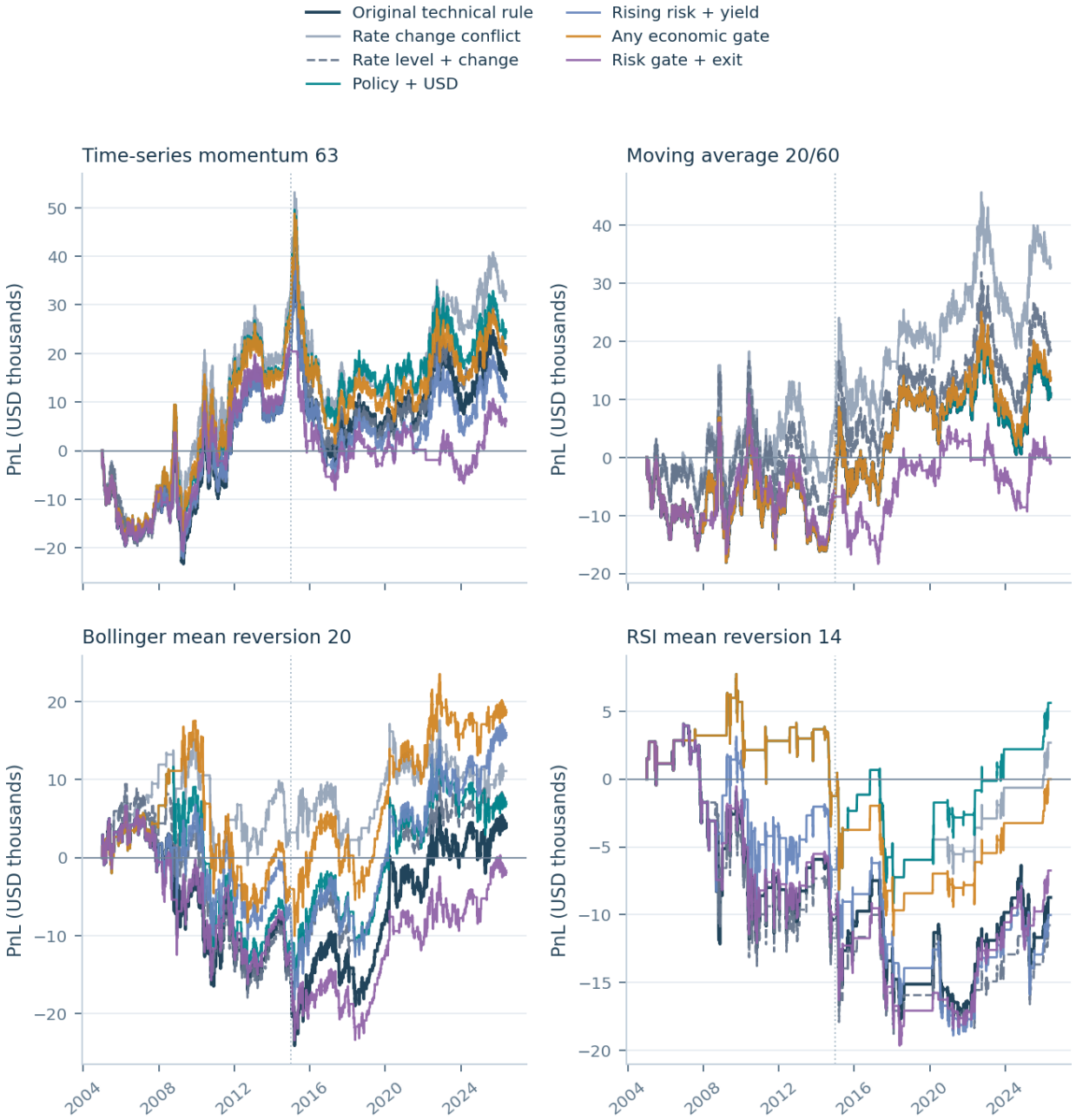
HTML explorer: open the self-contained file in a modern browser. Compare filters for a rule, strategies/windows for a currency, or currencies for a fixed rule. Change dates and inspect continuous PnL, return, drawdown and trade-cohort metrics. Separate sample selectors keep cash-reset and H.10 results distinct.

Datasets and code: use the inventory, dictionary and hashes to identify each source. The original files are unchanged. New code writes reconstructed results to a separate directory and does not overwrite the frozen evidence.

This final delivery adds no newly selected rule, parameter optimization or performance claim. It organizes and reproduces the work completed to date.

EUR | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

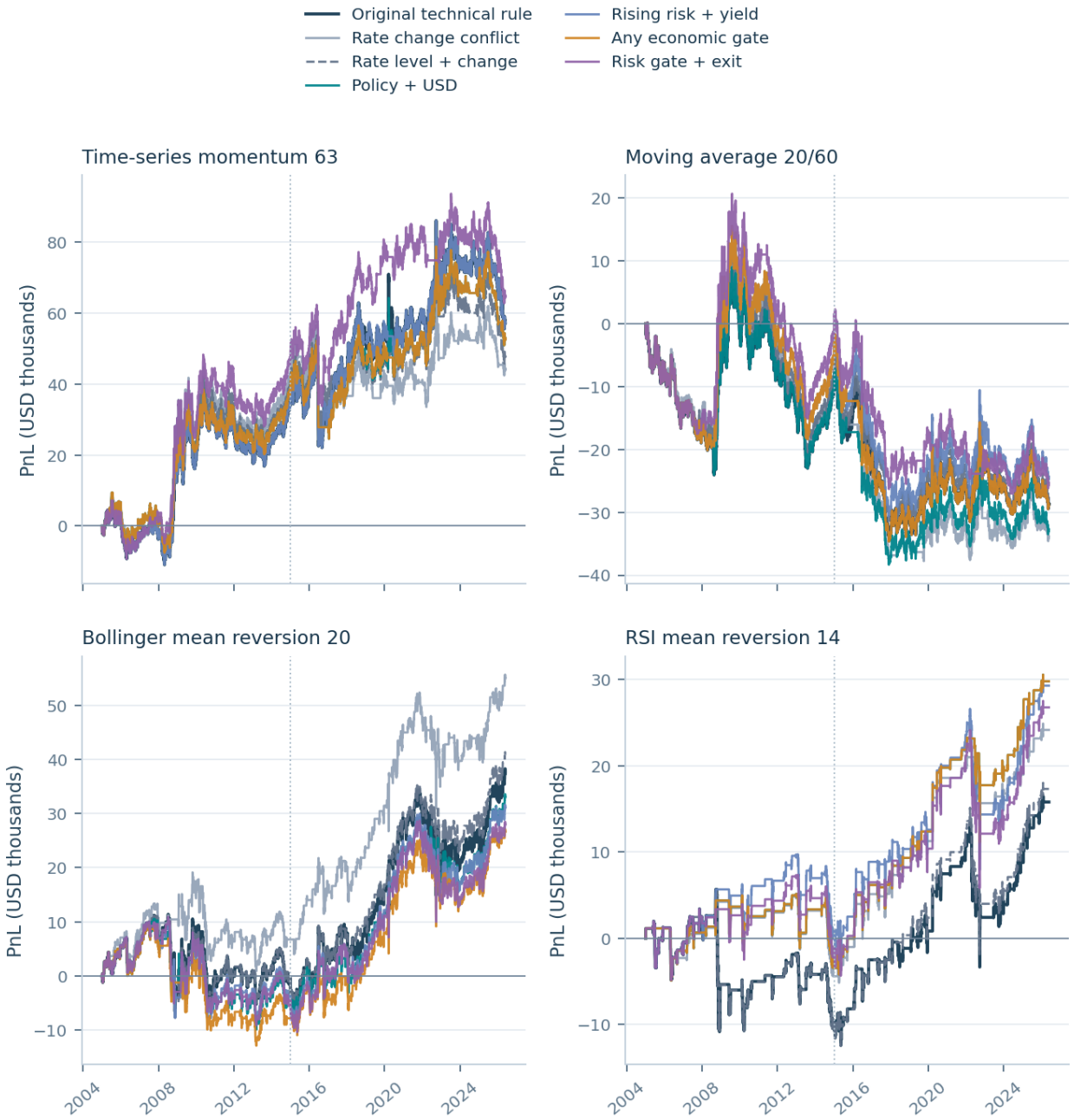


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	+15.77%	+20.85%	+6.04%
MA 20/60	+10.92%	+13.69%	-0.64%
Bollinger MR20	+4.33%	+18.89%	-1.70%
RSI MR14	-8.73%	-0.02%	-6.74%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

GBP | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

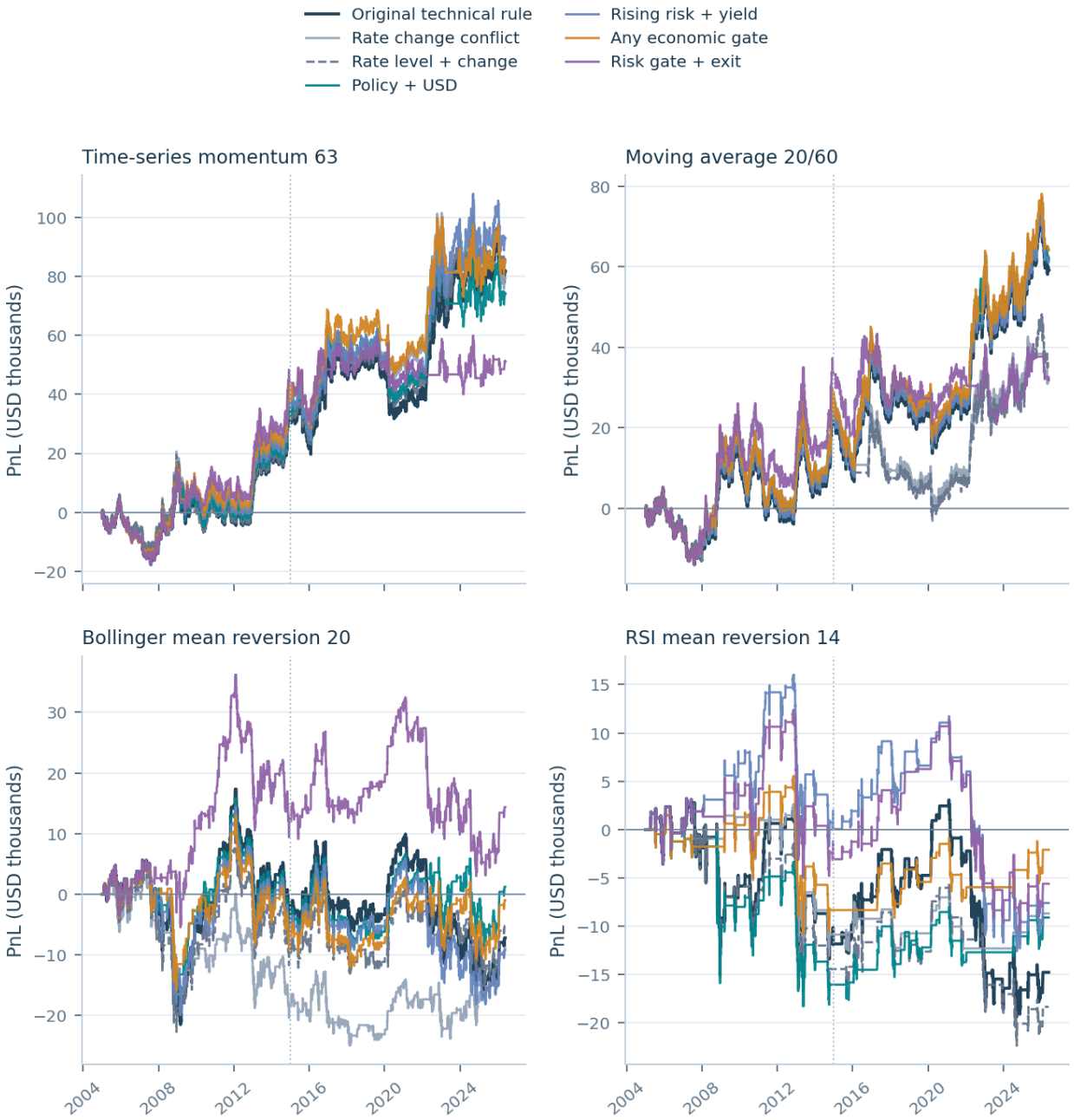


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	+57.40%	+52.75%	+64.72%
MA 20/60	-28.74%	-28.80%	-25.44%
Bollinger MR20	+37.67%	+26.59%	+27.92%
RSI MR14	+15.79%	+29.78%	+26.76%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

JPY | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

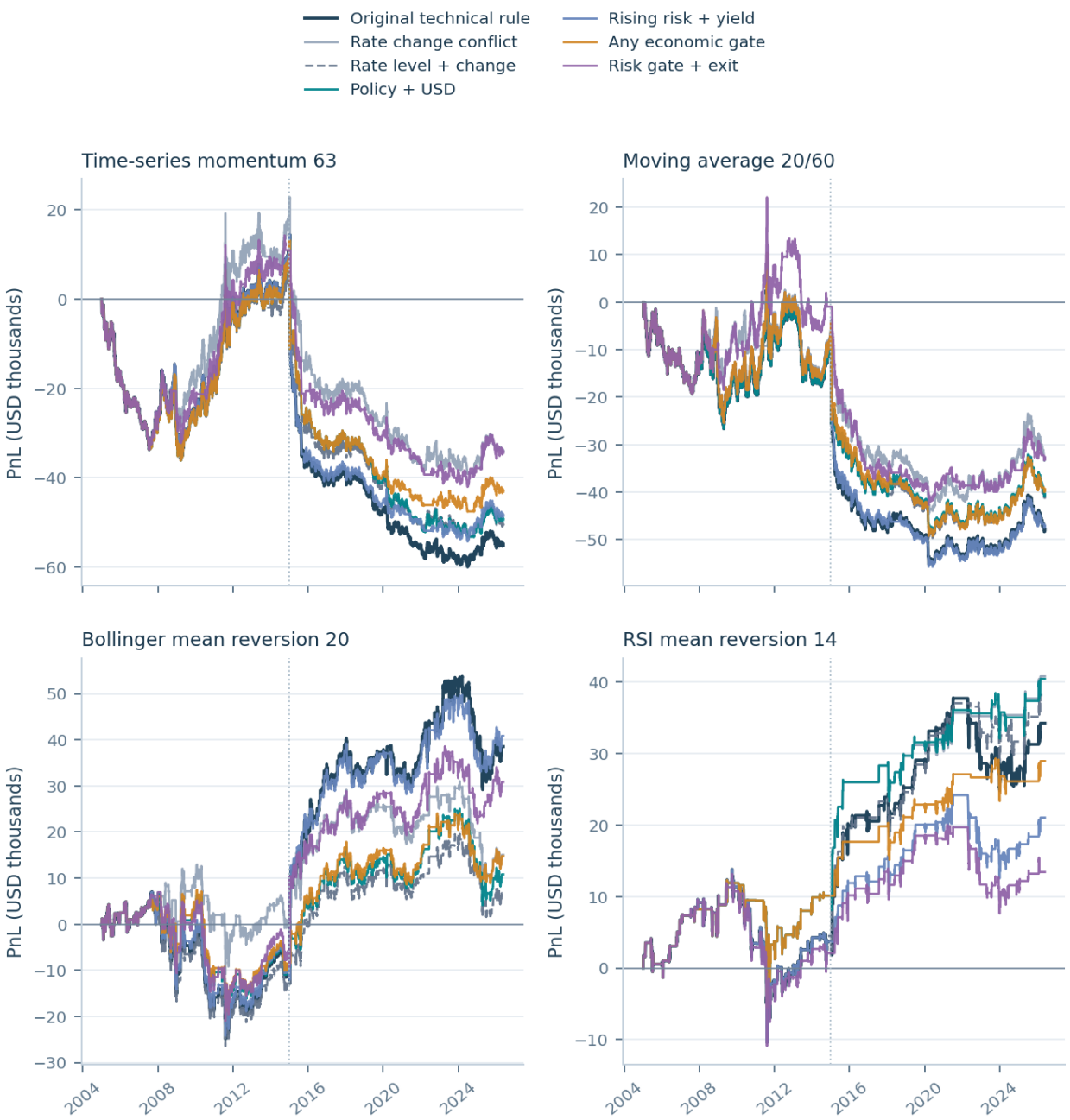


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	+81.78%	+84.92%	+51.17%
MA 20/60	+59.20%	+64.20%	+31.90%
Bollinger MR20	-7.23%	-1.00%	+14.36%
RSI MR14	-14.77%	-2.09%	-5.60%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

CHF | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

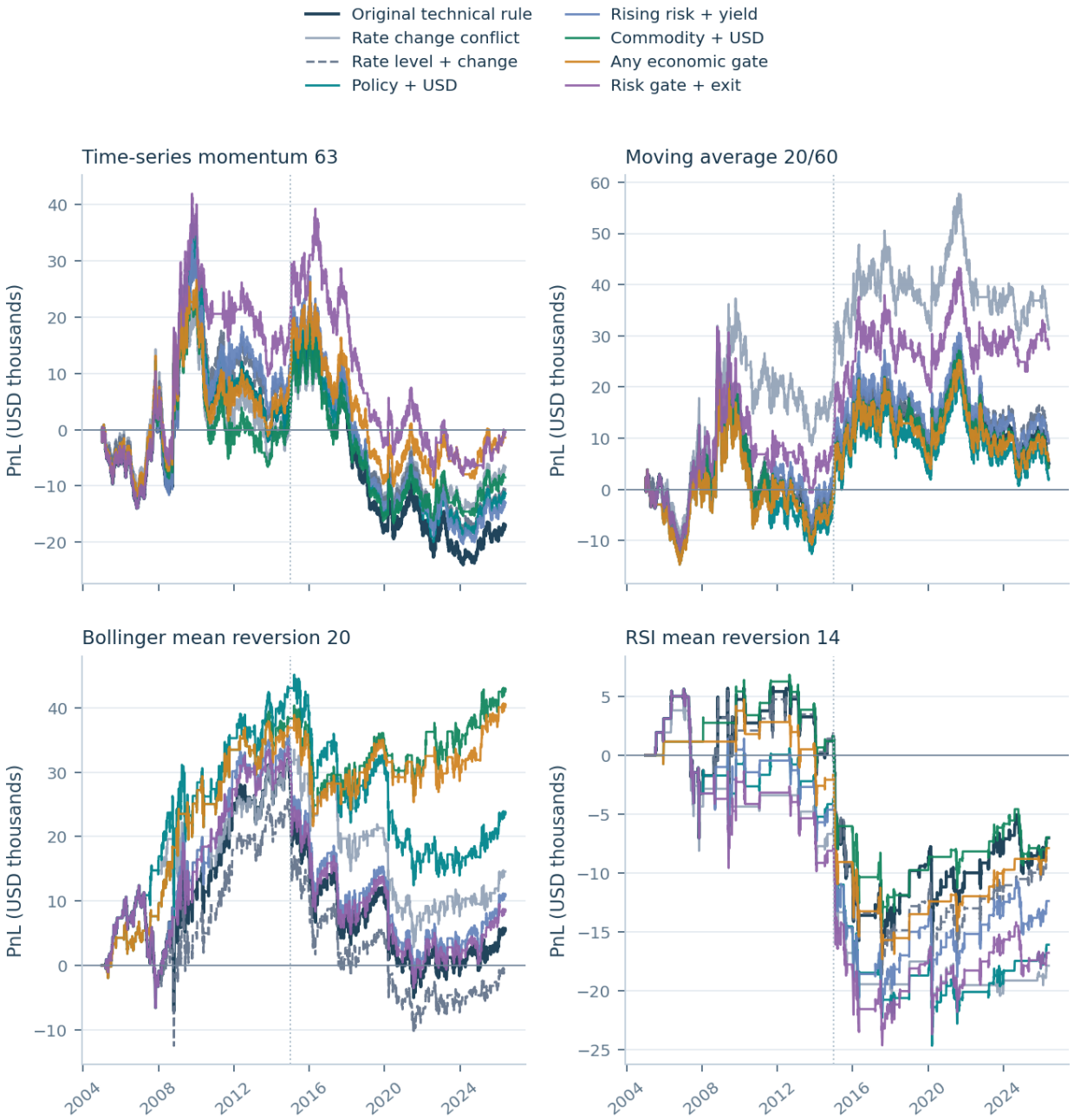


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	-55.09%	-43.11%	-34.43%
MA 20/60	-47.84%	-39.41%	-32.66%
Bollinger MR20	+38.53%	+14.75%	+30.82%
RSI MR14	+34.21%	+28.91%	+13.42%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

CAD | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

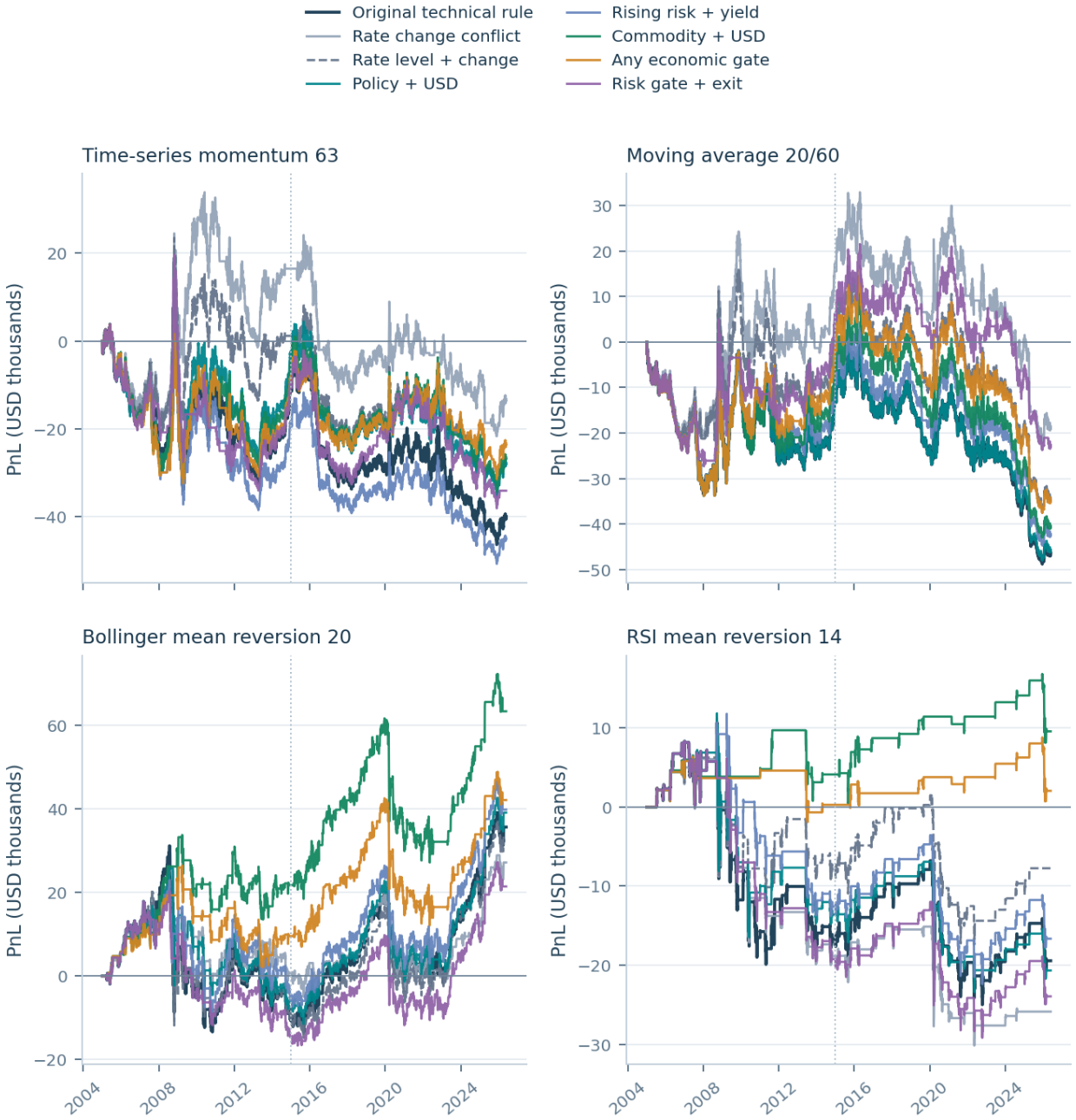


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	-17.15%	-1.43%	-0.51%
MA 20/60	+5.17%	+5.90%	+27.76%
Bollinger MR20	+5.72%	+40.56%	+8.68%
RSI MR14	-7.01%	-7.88%	-16.77%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

AUD | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

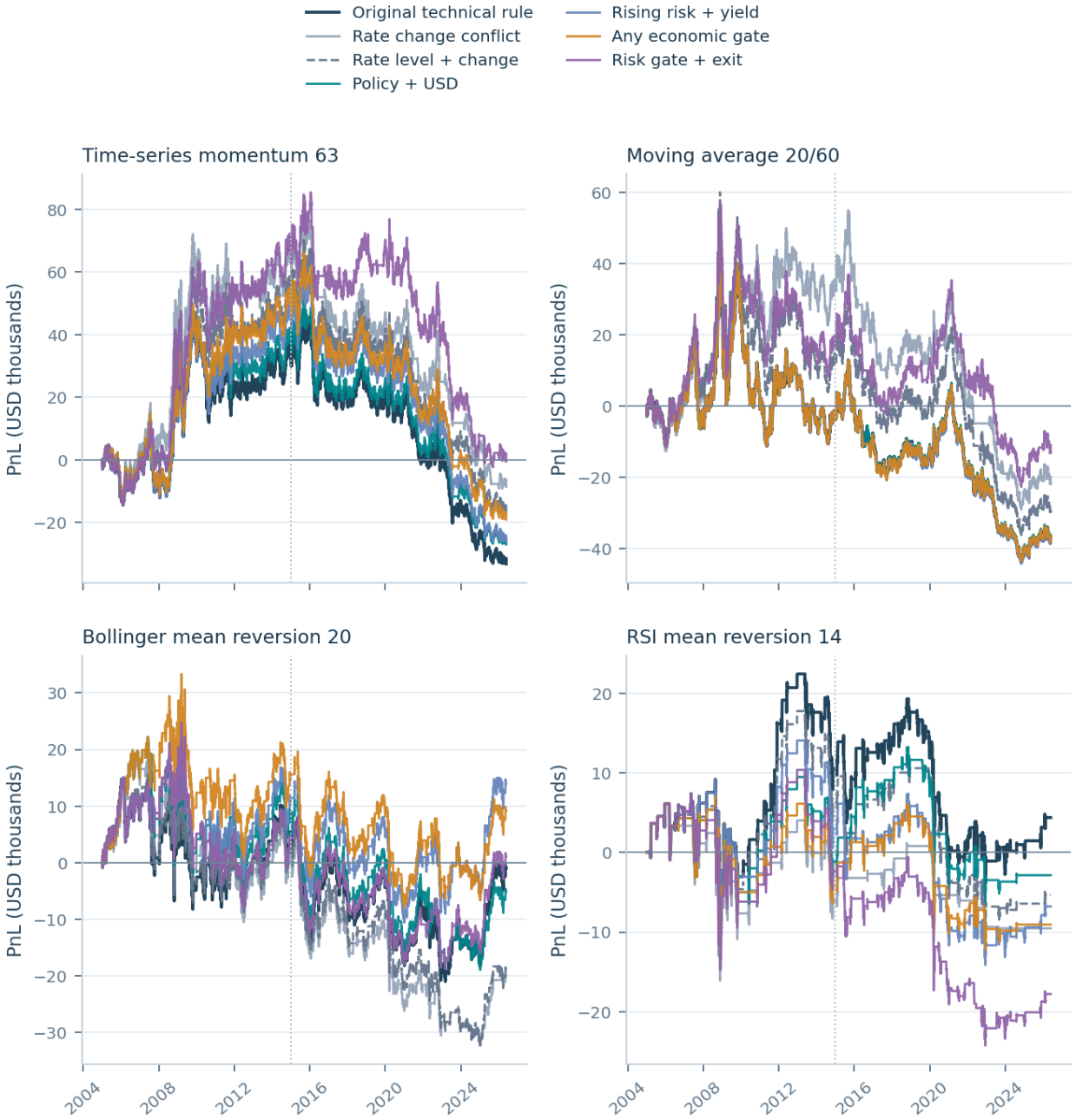


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	-39.98%	-23.31%	-34.10%
MA 20/60	-46.43%	-34.71%	-22.59%
Bollinger MR20	+35.71%	+42.18%	+21.47%
RSI MR14	-19.44%	+2.05%	-23.94%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

NZD | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

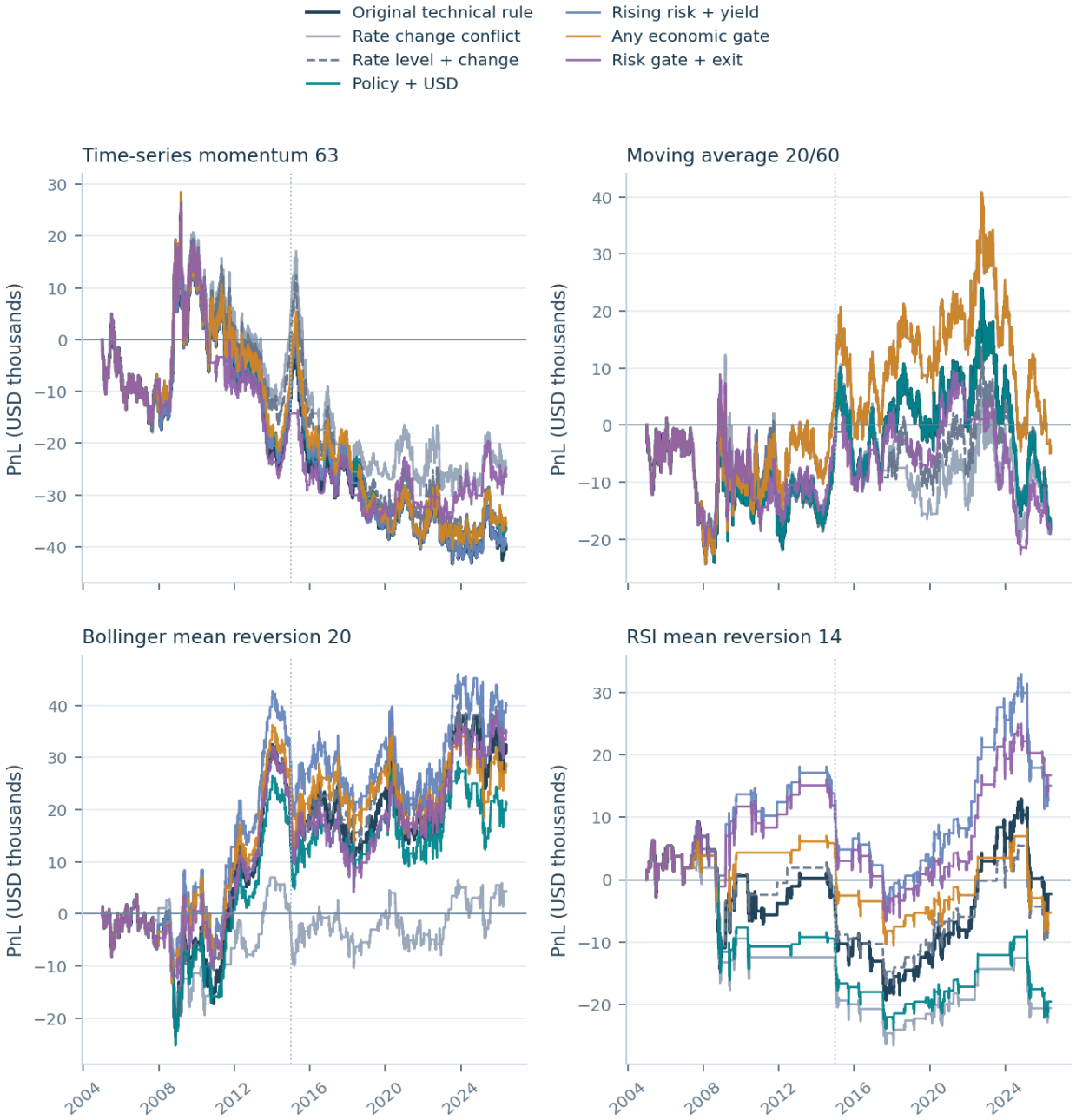


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	-33.20%	-19.04%	-0.52%
MA 20/60	-36.83%	-36.60%	-11.06%
Bollinger MR20	-0.99%	+9.17%	+1.05%
RSI MR14	+4.37%	-9.06%	-17.79%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

SEK | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.

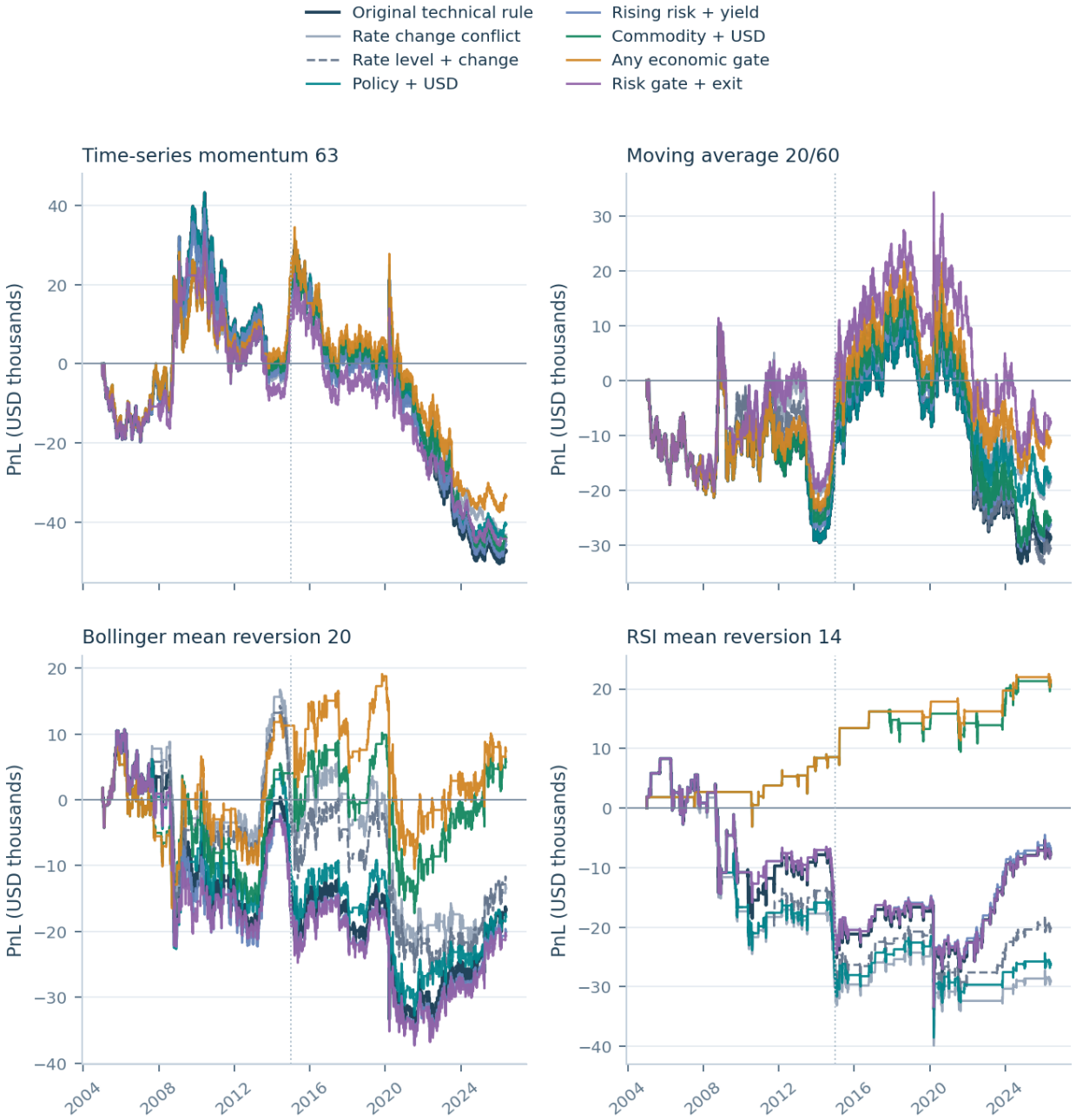


Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	-40.50%	-35.68%	-26.40%
MA 20/60	-18.18%	-4.89%	-19.02%
Bollinger MR20	+32.01%	+28.40%	+34.76%
RSI MR14	-2.27%	-5.30%	+16.73%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

NOK | Full-history filter atlas

Four fixed technical rules, independent USD 100,000 accounts; all valid economic-filter histories retained.



Rule	Original return	Any economic gate	Risk gate + exit
Momentum 63	-47.36%	-33.48%	-44.27%
MA 20/60	-28.67%	-11.07%	-7.62%
Bollinger MR20	-16.71%	+7.40%	-20.64%
RSI MR14	-7.70%	+21.20%	-7.79%

Continuous 2005-2026 gross spot PnL. Commodity/USD is shown only for AUD/CAD/NOK. The dashed 2015 marker is a historical review boundary; positions and wealth are not reset. The HTML provides exact daily hover values, selectable filters and all 20 baseline rules.

Primary sources | 1

External sources support mechanisms and data definitions. Internal numeric findings come from the supplied/frozen datasets.

[R1] Federal Reserve / Curcuru (2017)

The Sensitivity of the U.S. Dollar Exchange Rate to Changes in Monetary Policy Expectations [Open primary source]

Relative-policy expectations and changing exchange-rate sensitivity; does not validate the chosen cutoff.

[R2] Brunnermeier, Nagel & Pedersen (2009)

Carry Trades and Currency Crashes [Open primary source]

Carry crash and unwind mechanism; not a universal predictive VIX rule.

[R3] Bank of Canada (2017), Staff Analytical Note 2017-1

The Share of Systematic Variations in the Canadian Dollar - Part II [Open primary source]

Commodity and common-dollar influences on CAD; only the economic mechanism is borrowed.

[R4] RBA (2019)

Exploring the Supply and Demand Drivers of Commodity Prices [Open primary source]

Commodity demand and supply channels; copper is not a country-weighted export basket.

[R5] ICE

U.S. Dollar Index / USDX [Open primary source]

DXY basket construction and mechanical overlap with studied currencies.

[R6] Bank of Canada (2015-01-21)

Bank of Canada lowers overnight rate target to 3/4 per cent [Open primary source]

Oil-shock policy response after the CAD example entered; used only as historical context.

[R7] RBA (2021)

Determinants of the Australian Dollar Over Recent Years [Open primary source]

Australian dollar, relative rates, terms of trade and risk.

Primary sources | 2

External sources support mechanisms and data definitions. Internal numeric findings come from the supplied/frozen datasets.

[R8] Norges Bank (2022), Monetary Policy Report 4/2022

Section 2.3: The krone exchange rate [Open primary source]

Krone exchange rate, relative rates, terms of trade and risk/liquidity channels.

[R9] Sveriges Riksbank (March 2025)

Monetary Policy Report, pp. 8, 19 [Open primary source]

Krona and global developments; publication after the example exit.

[R10] Sveriges Riksbank (2025-03-19)

Minutes of the monetary policy meeting [Open primary source]

Dated discussion of financial flows and SEK; not a pre-February 2025 input.

[R11] BIS Bulletin 90 (2024-08-27)

The market turbulence and carry trade unwind of August 2024 [Open primary source]

August 2024 leveraged unwind and subsequent stabilization; retrospective mechanism.

[R12] BIS Annual Economic Report (2020), Chapter II

A monetary lifeline: central banks crisis response [Open primary source]

Dollar funding demand and crisis response in 2020; not identification of individual trade causality.

[T1] Neely, Weller and Ulrich / Federal Reserve Bank of St. Louis

The Adaptive Markets Hypothesis: Evidence from the Foreign Exchange Market (2007 working-paper revision; 2009 journal article) [Open primary source]

Previously published rule performance can change in later samples.

[T2] John Bollinger

Bollinger Bands Rules [Open primary source]

First-party indicator semantics, not evidence that the project strategy is profitable.

Primary sources | 3

External sources support mechanisms and data definitions. Internal numeric findings come from the supplied/frozen datasets.

[T3] Cboe Global Indices

Cboe Volatility Index Methodology [Open primary source]

VIX is 30-day US equity option-implied volatility; study thresholds are not official crisis labels.

[D1] Federal Reserve Board

About the H.10 Foreign Exchange Rates Release [Open primary source]

Quote time, weekly release, historical snapshots and revision distinctions.

[C1] EUR currency context

International funding and asset currency; a common policy area with heterogeneous sovereign yields. [Open primary source]

German yields proxy the euro area. Low rates do not by themselves identify unhedged carry positions or predict the euro-dollar direction.

Additional primary context: source 2

[C2] GBP currency context

International asset and liability positions and exposure to UK-specific policy/institutional uncertainty. [Open primary source]

An improved net international investment position through valuation is not a capital inflow or a reversal signal. A 2016 exposure estimate is not a permanent full-sample weight.

Additional primary context: source 2

[C3] AUD currency context

Commodity export income and terms of trade, interacting with relative rates and global risk. [Open primary source]

Copper is only a partial proxy, not an Australian export-weighted index. Demand shocks and commodity-specific supply shocks need not have the same currency effect.

Additional primary context: source 2

[C4] NZD currency context

Soft-commodity export income, including dairy, with relative rates and global risk exposure. [Open primary source]

The supplied data contain no dairy or country-weighted commodity index, so no commodity filter is forced onto NZD. 2008 rate coverage is only 11 of 262 dates; its annual median is not representative.

Additional primary context: source 2

Primary sources | 4

External sources support mechanisms and data definitions. Internal numeric findings come from the supplied/frozen datasets.

[C5] JPY currency context

A historically low-yield funding currency with large international financial exposures and position-adjustment channels. [Open primary source]

Do not assume repatriation or carry unwinds cause every appreciation. Policy regimes and execution timing change how a momentum or reversal signal behaves.

Additional primary context: source 2 | source 3

[C6] CHF currency context

Historically observed safe-haven exposure, conditioned by monetary intervention and exchange-rate arrangements. [Open primary source]

The floor was on EUR/CHF, not USD/CHF. Low global VIX cannot rule out a local policy discontinuity, and a safe-haven relation is not a permanent trading rule.

Additional primary context: source 2

[C7] CAD currency context

A diversified resource-exporting economy; commodity income and the common US-dollar factor both matter. [Open primary source]

CAD is not an oil contract. Brent and DXY are limited state proxies; they do not identify causal oil-supply versus global-demand shocks.

Additional primary context: source 2

[C8] NOK currency context

Energy income interacts with overseas assets, institutional hedging, fiscal currency conversion and market depth. [Open primary source]

Oil receipts are not all immediate NOK purchases. NOK 2Y coverage is absent in 2008 and has a long 2020 gap; do not invent a rate state during those intervals.

Additional primary context: source 2 | source 3

[C9] SEK currency context

Portfolio flows and relative policy expectations, with risk-premium sensitivity that changes across episodes. [Open primary source]

The higher-yield side can switch from SEK to USD. Later central-bank commentary is explanatory context, not information available at an earlier entry or exit.

Additional primary context: source 2